



Preliminary Modelling of the Ashburton Salt Proposal: Potential Future Changes to Habitat Extent



THE UNIVERSITY OF
WESTERN
AUSTRALIA

Document information summary

Item	Description		
Partner Name	Department of Water and Environmental Regulation		
Partner Contacts	Hans Kemps		
Date of issue	30/06/2025		
Report title	Preliminary Modelling of the Ashburton Salt Proposal: Potential Future Changes in Habitat Extent		
Prepared by	Daniel Raj David, Arnold van Rooijen, Ryan Lowe and Matthew Hipsey		
Report number	UWA-B-MS1175		
Version	<i>Date of issue</i>	<i>Prepared</i>	<i>Comments</i>
Draft A	14/4/2025	DRD	Submitted to HK, FW
Final	30/06/2025	DRD	Submitted to HK, FW

Contents

1. Summary	1
2. Introduction	2
2.1. Background	2
2.2. Study region	4
2.3. Scope and objectives	5
3. Hydrodynamic modelling	6
3.1. Mesh and setup	6
3.2. Simulation period and boundary conditions	7
3.3. Hydrodynamic model calibration	10
4. Habitat modelling	13
4.1. Habitat prediction validation	14
5. Scenario rationale	18
6. Future predictions	20
6.1. Baseline (Without Development)	20
6.2. With Development	21
6.3. Habitat migration	22
6.4. El Niño–Southern Oscillation (ENSO) effects	32
7. Limitations	36
8. Summary and conclusions	38
9. Appendix A	40
10. References	48

List of Figures

Figure 1 Proposed Project location in Western Australia (panel a), with a zoomed-in view of the Urala Creek focus site in panel b, showing the project boundary envelope marked in red.	4
Figure 2 Satellite derived key habitat classes in Urala based on 2023 data. Source: Map adopted from WAMSI Mardi Project A, see also Hickey et al., (2023) for a detailed methodology.	5
Figure 3 Delft3D Flexible Mesh full view in panel a) and the red box highlighted in panel a) is shown in panel b) a zoomed in view of the model mesh showing the variable resolution mesh to better capture the hydrodynamics.	7
Figure 4 Bathymetry of the full domain in panel a) and the red highlighted box is depicted in panel b) showing the zoomed-in view.	7
Figure 5 Boundary conditions for the representative year (2023).	9
Figure 6 Selected monitoring locations from which measured water level data were used for calibration purposes.	10
Figure 7 Measured and simulated water level at the Urala creek north mouth in left panel the location is highlighted in right. RBR is the water level measuring instrument.	11
Figure 8 Measured and simulated water level at the Urala creek north lower in left panel the location is highlighted in right. RBR is the water level measuring instrument.	11
Figure 9 Scatter plot of measured vs simulated (1 to 1) water level comparison for both Urala creek north mouth and lower. The red shaded area indicates the 25% error bar.	11
Figure 10 Daily averaged measured and simulated water level comparison at the Urala upper region.	12
Figure 11 Measured vs simulated water level at the Urala lower mat.	12
Figure 12 Simulated inundation extent during high tide on 14/02/2024 compared to the satellite observed extent on the same day.	12
Figure 13 Simulated inundation extent during low tide on 11/12/2023 compared to the satellite-observed extent on the same day.	13
Figure 14 Flowchart of the Random Forest (RF) approach illustrating the decision tree process and prediction, which is derived from the average of multiple decision trees. Image Source: Simplilearn.com.	14
Figure 15 Environmental factors and their distributions for different habitats based on hydrodynamic conditions in 2023.	15
Figure 16 Mapped and RF predicted habitat classes for the year 2021 and 2023 hydrodynamic conditions.	16
Figure 17 Confusion matrix (normalized) showing the accuracy of predictions.	17
Figure 18 Importance score as optimized in RF approach by the model for various hydrodynamic parameters.	18
Figure 19 Projected sea level rise based on average of SSP245, SSP370 and SSP585. The red dots and dotted lines indicate the adopted SLR scenarios for this study. SSP - Shared Socioeconomic Pathways.	19
Figure 20 Predicted habitat distribution under different SLR scenarios (see Table 3) for the baseline condition.	21
Figure 21 Predicted habitat classes and their spatial distributions under different SLR scenarios (see Table 3) for the development condition.	22
Figure 22 Mangrove distribution represented by gain and loss across different SLR scenarios compared to baseline(no SLR) without the development boundary envelope. The top panels	

show the overall domain, while the bottom panels show the zoomed-in view of the inset box shown in the top panels.	23
Figure 23 Cyanobacterial mat (algal mat) distribution represented by gain and loss across different SLR scenarios compared to baseline(no SLR) without the development boundary envelope. Top panels shows the overall domain while the bottom panels show the zoomed-in view of the inset box shown in the top panels..	24
Figure 24 Salt flat distribution represented by gain and loss across different SLR scenarios compared to baseline(no SLR) without the development boundary envelope. Top panels shows the overall domain while the bottom panels show the zoomed-in view of the inset box shown in the top panels..	25
Figure 25 Mangrove distribution represented by gain and loss across different SLR scenarios compared to the development baseline(no SLR), with the development boundary envelope highlighted in purple colour. The top panels show the overall domain, while the bottom panels show the zoomed-in view of the inset box shown in the top panels.	26
Figure 26 Cyanobacterial mat (Algal mat) distribution represented by gain and loss across different SLR scenarios compared to the development baseline(no SLR), with the development boundary envelope highlighted in purple colour. The top panels show the overall domain, while the bottom panels show the zoomed-in view of the inset box shown in the top panels.	27
Figure 27 Salt flat distribution represented by gain and loss across different SLR scenarios compared to the development baseline(no SLR), with the development boundary envelope highlighted in purple colour. The top panels show the overall domain, while the bottom panels show the zoomed-in view of the inset box shown in the top panels.	28
Figure 28 Predicted change in habitat area (in hectares, ha) without and with development footprint.	29
Figure 29 Future migration (all SLR scenarios combined) of habitats compared to the baseline scenario (no SLR).	31
Figure 30 Predicted future habitat extent compared to the baseline distribution. The figure highlights newly identified habitat areas under future conditions that were not classified as any habitat type in the baseline (present-day) distribution, indicating potential habitat expansion or shifts due to changing environmental conditions.	32
Figure 31 Decomposition of coastal sea level variability (Onslow, Western Australia), showing monthly maximum values for visual clarity. (a) Measured total water level (in blue) with monthly maximum highlighted in black lines along with extreme conditions (Tropical Cyclones, TC and Tropical Low, TL) highlighted with green vertical lines, and the decomposed (b) interannual and mean sea level trend contributions, (c) seasonal contribution, (d) surge contribution, and (e) tide contribution. (f) Southern Oscillation Index (SOI), with highlights for values ± 1 indicating an increasing likelihood of El Niño and La Niña events.	33
Figure 32 Difference in the percentage of submergence based on ENSO positive, negative, compared to ENSO neutral conditions.	35
Figure 33 Inundation extent for ENSO positive and negative scenarios.	35
 Figure A 1 Mean water depth across different SLR scenarios without the development boundary.	 40
Figure A 2 Mean water depth across different SLR scenarios with the development boundary.	40
Figure A 3 Maximum depth across different SLR scenarios without the development boundary.	41
Figure A 4 Maximum depth across different SLR scenarios with the development boundary.	41

Figure A 5 Mean current magnitude across different SLR scenarios without the development boundary	42
Figure A 6 Mean current magnitude across different SLR scenarios with the development boundary.	42
Figure A 7 Maximum current magnitude across different SLR scenarios without the development boundary.	43
Figure A 8 Maximum current magnitude across different SLR scenarios with the development boundary.	43
Figure A 9 Percentage of submergence based on 1-year simulation for different SLR scenarios without the development boundary.	44
Figure A 10 Percentage of submergence based on 1-year simulation for different SLR scenarios with the development boundary.	44
Figure A 11 Number of dry days for different SLR scenarios without the development boundary.	45
Figure A 12 Number of dry days for different SLR scenarios with the development boundary.	45
Figure A 13 Number of wet days across different SLR without the development boundary.	46
Figure A 14 Number of wet days across different SLR with the development boundary.	46
Figure A 15 Maximum consecutive wet days across different SLR scenarios without the development boundary.	47
Figure A 16 Maximum consecutive wet days across different SLR scenarios with the development boundary.	47

List of Tables

Table 1 Modelling period	8
Table 2 Roughness values	8
Table 3 Scenarios used for the habitat predictions.	20
Table 4 Predicted change in habitat area (in hectares, ha) for different SLR scenarios without and with development footprint.	30

1. Summary

As part of Project B of the WAMSI Mardie Intertidal Research Offset Program, models have been developed that can explore the potential impacts of projected future sea level rise (SLR) and hydrodynamic variability on intertidal habitats in the Pilbara coastal region of Western Australia. This includes the development of a Delft3D Flexible Mesh (DFM) hydrodynamic model, combined with a machine learning-based habitat prediction model, to examine how key habitats—such as mangroves, cyanobacterial mats, and salt flats—may respond to long-term changes in inundation regimes. This technical report describes work undertaken to use this modelling system to the Urala Creek region of Exmouth Gulf to explore the sensitivity of habitat change to the presence of salt-pond development footprint that is proposed for construction.

Various scenarios were assessed, including different SLR projections (2050, 2075, 2100), the influence of ENSO-related sea level anomalies, with and without the impact of development boundaries such as bund walls. The habitat prediction model, based on a Random Forest classification approach, used hydrodynamic parameters like submergence/exposure duration, water depth and depth-averaged currents to estimate the spatial distribution and migration of habitats under these scenarios. The model is simplified in that it assumes a fixed bathymetry, and thus does not account for potential morphodynamic feedbacks that could occur as habitats migrate; nonetheless, it can demonstrate how the nature of changes to coastal inundation patterns will likely drive changes in future habitat extent.

Results suggest that as sea levels rise, intertidal habitats tend to migrate inland in response to the changing inundation conditions. Cyanobacterial mats were found to be the most dynamic, with the model predicting potentially significant inland shifts and expansion up to a depth threshold, beyond which their extent declines. Mangroves remained relatively stable under moderate SLR but exhibited noticeable retreat and loss under extreme conditions projected for 2100. Salt flats and high intertidal flats emerged in newly inundated areas, suggesting shifting zonation patterns.

The analysis indicates that portions of the development boundary, particularly within salt production areas, may become increasingly suitable (yet unavailable) for habitat colonisation, especially by cyanobacterial mats. A comparison of results outside the boundary envelope for scenarios with and without the salt farm bund walls revealed only marginal differences. This suggests that the presence of the bund walls did not significantly alter the environmental conditions influencing habitat distribution outside the boundary envelope. The influence of ENSO variability further emphasised the sensitivity of specific habitats, particularly cyanobacterial mats, to short-term fluctuations in water levels.

Overall, this integrated modelling approach offers an initial assessment for anticipating habitat change under future scenarios, supporting proactive environmental planning and management. While not intended to provide exact future distributions, the insights gained can guide regulators and stakeholders in assessing ecological risks and formulating adaptive strategies.

2. Introduction

In recent years, there has been a growing interest in establishing salt farms along the northwest coast of Western Australia. The region's unique coastal characteristics, including its significant tidal range and the presence of naturally occurring salt flats, have made it an ideal location for solar-based salt production. Since 1967, numerous salt projects have been proposed, with many receiving approval and several already constructed. Given the increasing number of proposals for new large-scale salt ponds, the Department of Water and Environmental Regulation has sought expert analysis to better understand the potential environmental impacts. As part of this effort, the University of Western Australia has been contracted to conduct a modelling assessment of the Ashburton Salt Proposal, focusing on its potential influence on habitat extent and evolution.

2.1. Background

The Pilbara region in Western Australia is home to unique intertidal ecosystems, where extensive tidal creeks, salt flats, and low-lying coastal landforms create a complex eco-geomorphic setting. These arid-zone coastal wetlands are rare globally, as they are shaped by intricate interactions between salinity, sediment transport, and habitat zonation, which are in turn influenced by tidal cycles. Episodic flooding events and sea level variability further contribute to a dynamic inundation regime, redistributing salt, nutrients, and sediments across the intertidal landscape. However, despite the ecological significance of these environments, there is a notable lack of data and understanding regarding their behaviour, making it challenging for regulators to assess the impacts of environmental changes and coastal infrastructure.

The intertidal areas of the West Pilbara Coast exhibit a distinct zonation of mangrove, samphire, and algal (cyanobacterial) mat communities, transitioning from subtidal embayments to terrestrial margins. These habitats are governed by complex ecohydrological interactions, including tidal inundation, evaporation, rainfall-driven freshwater inputs, and exchange between surface and groundwater. The extent to which these ecosystems support surrounding marine and terrestrial environments remains unclear. Additionally, given their sensitivity to changes in salinity and inundation, any abrupt environmental shift—whether caused by climate change or human development—may trigger habitat loss or redistribution. For instance, minor sea level anomalies observed during the 2015/16 El Niño event led to significant mangrove dieback across northern Australia (Lovelock et al., 2017), demonstrating the vulnerability of these ecosystems to even small environmental perturbations.

Physical and ecological processes interact to determine how intertidal vegetation and habitats respond to changes in sea level (Wimmler et al., 2021). Different plant communities exhibit varying capacities to attenuate flows and influence local sediment transport, affecting coastal erosion and accretion over time. These processes can play an important role in shaping coastal morphology, influencing the resilience of intertidal environments against coastal hazards, including extreme storms and rising sea levels. For example, the loss of vegetation, particularly mangroves, has been directly linked to accelerated erosion in parts of Southeast Asia (Mazda et al., 2002). Superimposed on these natural dynamics are anthropogenic stressors, including coastal infrastructure developments and resource extraction, which may exacerbate habitat vulnerability and reduce the ability of these ecosystems to adapt to future environmental changes.

While previous conceptual and numerical models have provided insight into the functioning of these ecosystems, a critical gap remains in understanding their sensitivity to both anthropogenic and climatic pressures. A comprehensive approach is needed to assess how these habitats will evolve under future conditions and to determine the capacity of intertidal ecosystems to migrate and persist in response to shifting environmental baselines. Several processes are involved in predicting habitat evolution or migration, including sediment transport dynamics, changes in inundation regimes, temperature, salinity, and feedback between vegetation and hydrodynamic forces. However, to predict these changes with a certain level of accuracy, a fully coupled dynamic model is required—one that integrates hydrodynamics, sediment transport, and ecological responses. Currently, no models exist for this area that can dynamically resolve these complex interactions.

Several studies have relied on purely hydrodynamic models to characterize inundation frequency and predict the migration of habitats, particularly for mangroves (Adame et al., 2021; Cannon et al., 2020; Henderson & Glamore, 2024; Kumbier et al., 2021; Xie et al., 2022). While this principle holds true for other habitats, such as algal mats (also referred to as cyanobacterial mats), there is a lack of well-defined thresholds that can be readily applied to model their response to environmental changes. Consequently, predicting habitat changes due to sea level rise and coastal developments, such as salt farms, requires a more advanced modelling approach that incorporates both hydrodynamic and ecological factors.

To predict these intertidal habitats migration, in this analysis, we used a combination of hydrodynamic modeling and data-driven approaches to simulate both the hydrodynamic processes and the potential spatial extent of habitats within the Urala Creek system and its associated tidal salt flat under both the current conditions and a range of future development scenarios (see [Section 4](#) for detailed habitat modelling). The assessment integrates two key models: (1) a Delft3D-FM Hydrodynamic Model (Kernkamp et al., 2011), which has been recently developed for the Urala Creek system as part of the WAMSI Mardie Offset Research Project B and has been validated for water level and inundation extents; and (2) the Urala Creek Habitat Suitability Model, a data-driven model that predicts habitat locations based on historical inundation patterns. The latter is a novel approach that is still under development, incorporating measures of model accuracy and uncertainty to enhance predictive reliability.

By running multiple scenarios, this report aims to evaluate the sensitivity of habitat changes under combined pressures of sea level rise and salt-pond development. Given the complexity and uncertainty associated with habitat evolution processes, the model should not be viewed as providing an exact prediction of future habitat extent. Instead, it aims to indicate whether habitats are likely to persist under future scenarios and where we can most anticipate habitat migration based on the best available estimates of future inundation regimes. Unlike existing studies, such as Seashore Engineering (2022), which rely on conceptual models and heuristic forecasts of habitat change under SLR, our approach utilises process-based hydrodynamic modelling to simulate spatially explicit changes in inundation patterns. This enables a more detailed understanding of hydrodynamics in habitat zones, which may shift in response to dynamic water level variations, rather than assuming broad-scale migration trajectories. While our model still includes some assumptions (e.g., static bathymetry, see also [Section 7](#) for more limitations), it offers a more spatially resolved and quantitative foundation for evaluating future habitat distributions.

This work will contribute valuable insights into the resilience of intertidal habitats in the West Pilbara region and inform strategies for sustainable coastal development and conservation.

2.2. Study region

Figure 1 shows the location of the study site in Western Australia (Figure 1a), with the proposed development boundary envelope marked in Figure 1b. The Urala Creek system is a distinctive coastal environment characterised by two wide creeks — Urala Creek North and Urala Creek South — that approach from opposite directions and interconnect through a network of shallow and narrow creeks. These interconnected waterways form a dynamic hydrological system that plays a critical role in shaping the region's intertidal and supratidal habitats. For consistency with existing reports, these creeks are referred to as Urala Creek North and Urala Creek South, respectively.

The tidal flats associated with Urala Creek North and South generally have elevations ranging from 0.2 to 1.0 m AHD, creating a low-gradient intertidal and supratidal environment. This system supports a diverse array of habitats, including cyanobacterial mats, mangroves, salt marsh, samphires, coastal vegetation, and extensive salt flats. These habitats are acknowledged to serve important ecological functions within the region, contributing to nutrient cycling, sediment stabilisation, and coastal biodiversity. For this project, our focus is primarily on four habitat classes: cyanobacterial mat, mangrove, high intertidal flat, and salt flat.

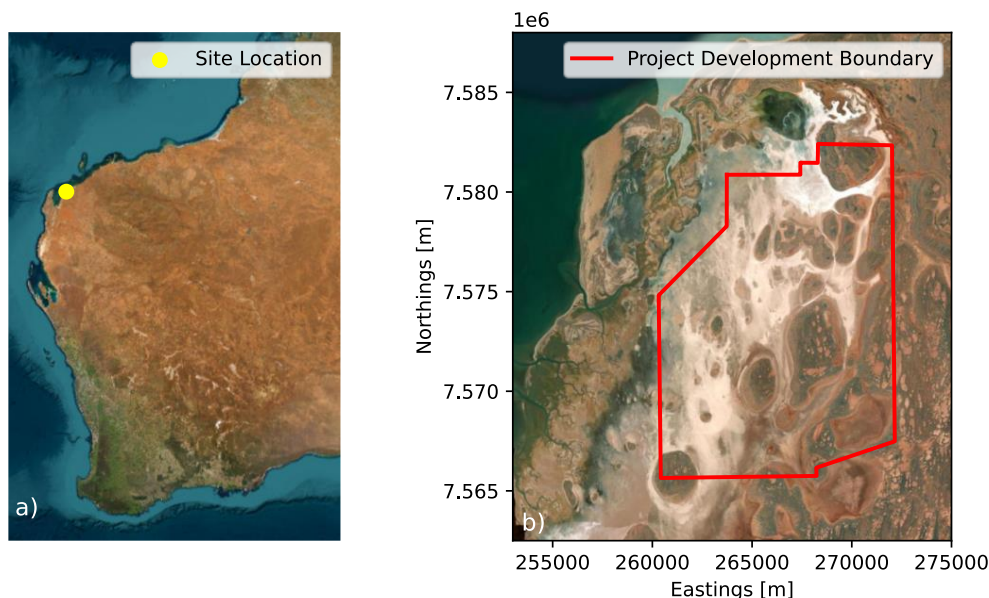


Figure 1 Proposed Project location in Western Australia (panel a), with a zoomed-in view of the Urala Creek focus site in panel b, showing the project boundary envelope marked in red.

Figure 2 presents a satellite-derived habitat map (based on 2023 data, Hickey et al., 2023) developed as part of the Mardie Project A, highlighting the spatial distribution of these key coastal habitats. The mangrove, algal mat, and salt flat zones within the Urala Creek system experience cyclical tidal inundation. Water levels primarily dictate the extent of flooding, with significant inundation occurring when water levels exceed Mean High Water Springs (MHWS) (~0.8 m AHD).

Under these conditions, tidally induced shallow sheet flows submerge expansive intertidal areas, particularly around Urala Creek North.

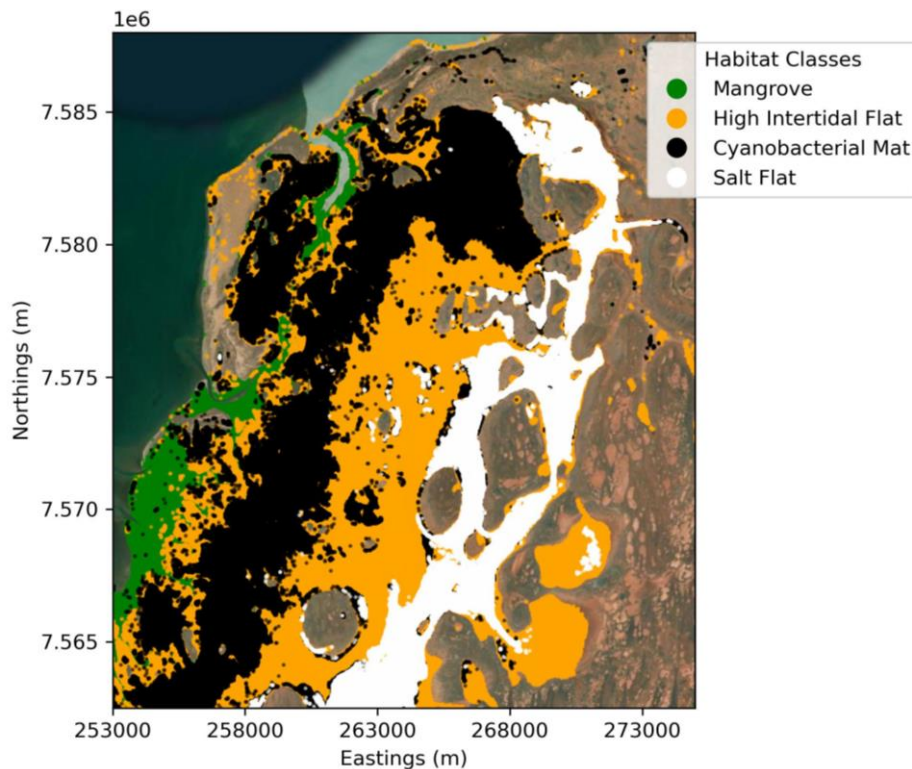


Figure 2 Satellite derived key habitat classes in Urala based on 2023 data. Source: Map adopted from WAMSI Mardi Project A, see also Hickey et al., (2023) for a detailed methodology.

This hydrodynamically complex system is potentially sensitive to environmental changes, including sea level rise and coastal development. The interaction between tides, sedimentation, and vegetation zonation makes Urala Creek an interesting natural setting for exploring intertidal habitat evolution and resilience in response to shifting coastal processes.

2.3. Scope and objectives

This project aims to assess the hydrodynamics and habitat spatial evolution of the Urala Creek system under current and future conditions, and with and without future development present. The study focuses on four key intertidal habitats: cyanobacterial mats, mangroves, high intertidal flats, and salt flats, which are influenced by changes in inundation patterns and sea level variations.

A combination of hydrodynamic and data-driven modelling was used. The Delft3D-FM hydrodynamic model, developed as part of the WAMSI Mardie Offset Research Project B, simulates water levels and currents, while a data-driven habitat prediction model is used to estimate habitat migration based on historical inundation regimes.

A matrix of scenarios is considered, including:

- Projected future sea level rise time points (2050, 2075, 2100)

- Sensitivity to periodic sea level anomalies (e.g., El Niño)
- Impacts of bund walls associated with salt-pond development

While the model does not aim to provide precise future habitat distributions, it will offer insights into potential changes to the hydrological niche for habitats, aiding regulatory agencies and stakeholders in environmental planning and management.

3. Hydrodynamic modelling

3.1. Mesh and setup

The grid design for the Urala domain covered an area of approximately 44 km in length and 35 km in width. The grid consisted entirely of triangular elements with variable resolutions, ranging from 250 meters in open water areas to 12 meters near shorelines and key locations such as creek mouths. This high-resolution approach ensured accurate representation of hydrodynamic processes in the study area. Figure 3a provides an overview of the model domain, while Figure 3b presents a zoomed-in view of the variable resolution mesh within the highlighted red box, demonstrating finer grid spacing in critical regions. As the focus of this study is to predict habitat migration, optimising the domain extent is a key consideration. To achieve this, we arbitrarily considered an extreme SLR scenario of +1.5 m, simulated for one year, and then fixed the domain extent based on the results. The adopted extent was deemed appropriate for the objectives of this study.

The bathymetry and topography incorporated into the model were derived from multiple high-quality sources, including:

- **LiDAR topography** from surveys conducted by the University of Western Australia (UWA);
- **Creek bathymetry surveys** conducted by UWA as part of Project A (existing data);
- **Satellite-derived bathymetry** (from Geoscience Australia) data for the Exmouth Gulf.

These combined datasets facilitated the development of a comprehensive and high-resolution bathymetric map. Figure 4a shows the bathymetry of the model domain, while Figure 4b presents a zoomed-in view of the Urala North and South creeks, along with intertidal flats. Multiple iterations were performed, adjusting the bathymetry based on comparisons with observed water level data to enhance model accuracy. Specific refinements were made at critical locations, such as the northern and southern mouths of Urala Creek, to better align the simulated hydrodynamics with measured conditions.

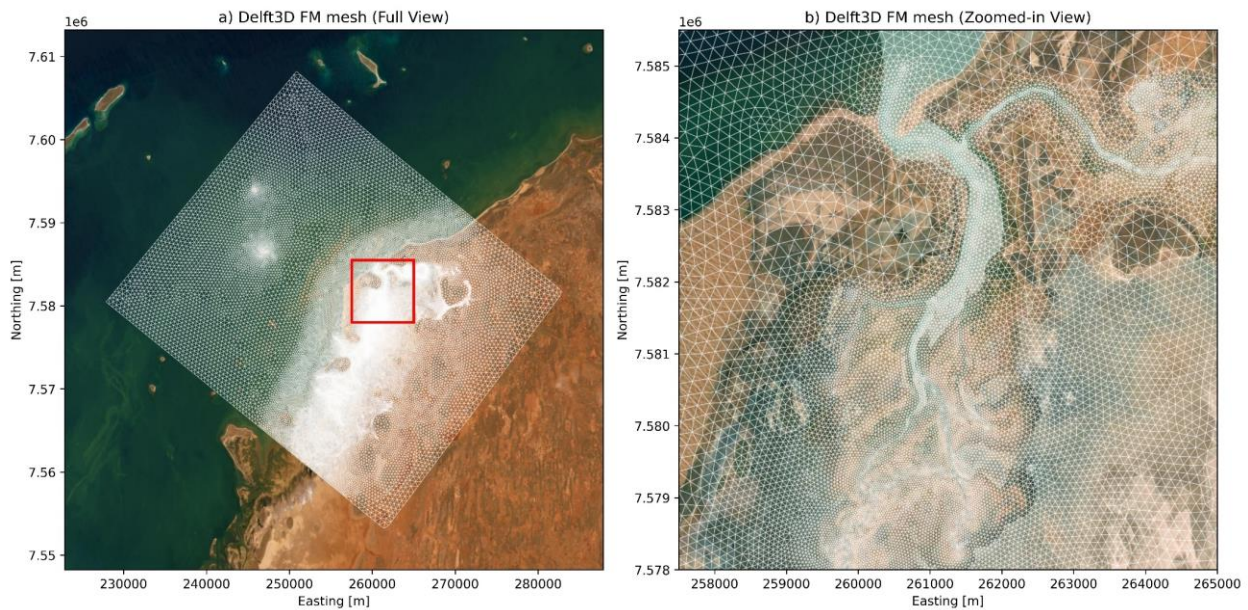


Figure 3 Delft3D Flexible Mesh full view in panel a) and the red box highlighted in panel a) is shown in panel b) a zoomed in view of the model mesh showing the variable resolution mesh to better capture the hydrodynamics.

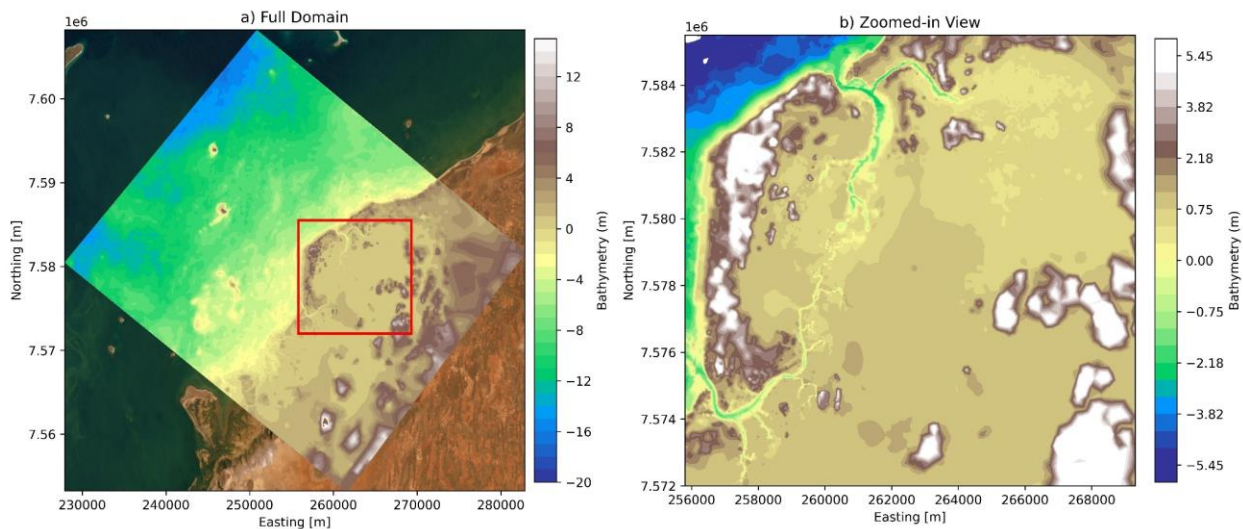


Figure 4 Bathymetry of the full domain in panel a) and the red highlighted box is depicted in panel b) showing the zoomed-in view.

3.2. Simulation period and boundary conditions

The modelling period for this study was determined based on the availability of hydrodynamic validation data and satellite-derived habitat mapping. Field-collected data from a larger field campaign, conducted between November 2023 and mid-April 2024 (Phase 1 – the first field-measured period), were used for model calibration and validation. To ensure a representative period for hydrodynamic analysis, we selected the full year of 2023 for further simulations (for habitat predictions). This period encompasses part of the calibration and validation phase (November 2023 to April 2024) and aligns with the satellite-derived habitat map developed as part of Project A. Table 1 shows the modelling period.

Table 1 Modelling period

Modelling task	Simulation period
Calibration	November 2023 to April 2024
Representative year	January 2023 to December 2023

For hydrodynamic simulations, we incorporated key physical processes and forcing conditions relevant to this study, including tidal generating forces, evaporation (heat flux model) with inputs of air temperature, cloud cover and humidity, meteorological forcing such as wind, bottom roughness, and rainfall. Rainfall was included only for the calibration period, whereas for the predictive simulations assessing habitat-specific hydrodynamic conditions, we considered only tidal water levels. The primary water level inputs were obtained from the Onslow tide gauge station (Western Australia), while other atmospheric and oceanic inputs were sourced from the ERA5 reanalysis dataset. In this study, we allowed Delft3D FM to calculate evaporation as part of the composite heat flux approach, which accounts for net surface heat exchange based on input meteorological data. The model utilised ERA5 Reanalysis data (including cloud cover, relative humidity, air temperature, and wind speed) to compute the heat flux components. The solar radiation is computed internally based upon time of day and cloudiness. The composite heat flux method in Delft3D FM estimates evaporation through bulk aerodynamic formulations, where latent heat flux (and hence evaporation) depends on the difference between surface and atmospheric humidity, wind speed, and atmospheric pressure, among other factors. This method offers a physically consistent way to simulate heat and moisture fluxes at the water–air interface, avoiding the need for user-defined or simplified evaporation inputs.

Table 2 Roughness values

Habitat/Topographic type	Manning's roughness coefficient
Cyanobacterial Mat	0.03
Mangrove	0.12
Water and offshore region	0.023
Salt Flat	0.045
High Intertidal Flat	0.05
Unvegetated Sand	0.06

Figure 5 shows the boundary conditions used for the representative year simulation, including water levels, humidity, air temperature, cloud cover, and winds. Seasonal variations are evident in the water level panel, where the red line represents the monthly mean. During the initial calibration phase, additional water level inputs were tested, including outputs from the TPXO

global tide model and HYCOM for water level anomalies. However, statistical comparisons using metrics such as Root Mean Square Error (RMSE), bias, scatter index, and slope indicated that measured water levels provided the most representative forcing. The measured data inherently captured extreme events, such as storm surges, which are critical for habitat dynamics.

To account for flow resistance across different habitat types, we used the 2023 mapped habitat distributions and assigned roughness values based on Manning's coefficients. Table 2 provides the adopted Manning's coefficient values for each habitat type.

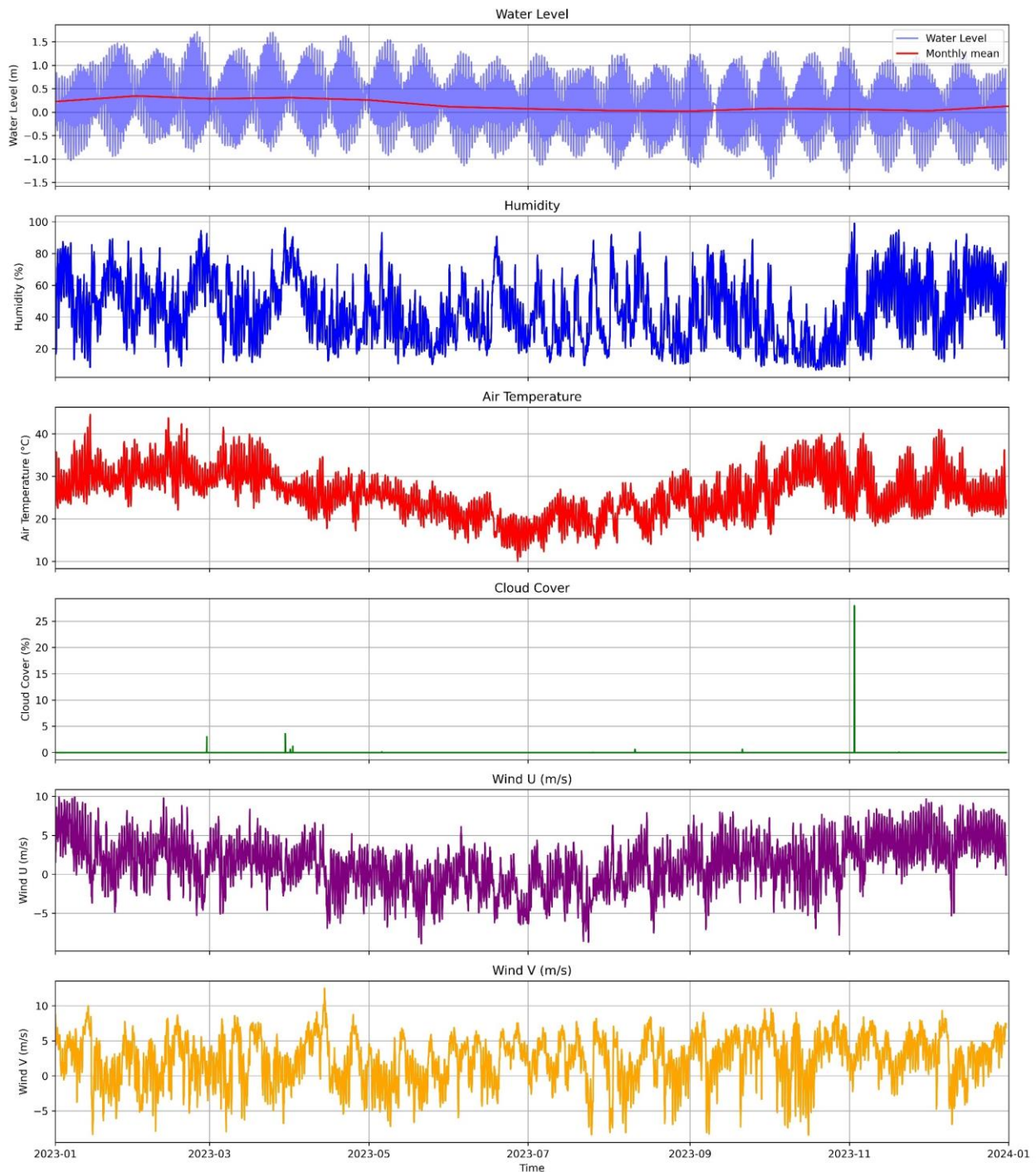


Figure 5 Boundary conditions for the representative year (2023).

3.3. Hydrodynamic model calibration

The hydrodynamic model calibration was carried out using measured water levels at selected monitoring locations (see Figure 6 for locations), including the Urala Creek North mouth, lower creek, and both lower and upper algal mats. These locations were chosen to capture the key hydrodynamic characteristics across different habitat zones. Although salinity and temperature are important factors influencing habitat migration, the primary driver in this study is water level. Due to inherent uncertainties in the simulated salinity and temperature data, our habitat migration predictions are based solely on hydrodynamic conditions (i.e., inundation and flow parameters).

Figure 7 to 11 present a comparison between measured and simulated water levels at the selected locations. The results indicate a reasonable agreement between the two datasets. To quantify the model's performance, we computed several statistical metrics, including root mean square error (RMSE), bias, slope, scatter index, and correlation coefficient. At the mouth of Urala Creek and the lower creek, the model exhibited excellent agreement with measured data, with an RMSE of approximately 0.068 m, a bias of -0.07 m, a slope of 0.99, a scatter index of around 0.17, and a correlation coefficient of approximately 0.98.

The comparison of surface water levels at the lower and upper algal mats showed moderate accuracy. However, when analysing the amplitude and frequency of inundation, the model demonstrated a reasonable performance, capturing approximately 70% of the observed inundation events (Figure 10 and 11). To further validate the model, we compared the simulated inundation extent with satellite-derived water coverage during a representative high tide and low tide within the study period. The spatial comparison showed a good match (Figure 12 and 13), reinforcing confidence in the model's ability to simulate hydrodynamic processes accurately. Given this reliability, the hydrodynamic characteristics extracted for different habitat zones provide a basic foundation for predicting habitat changes under future SLR scenarios.

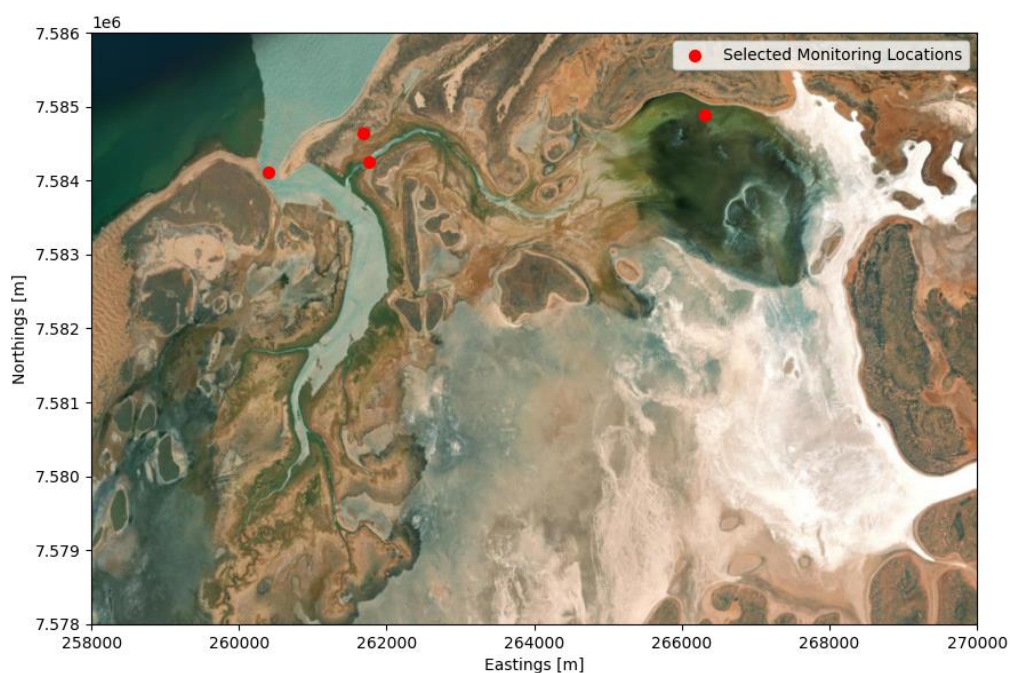


Figure 6 Selected monitoring locations from which measured water level data were used for calibration purposes.

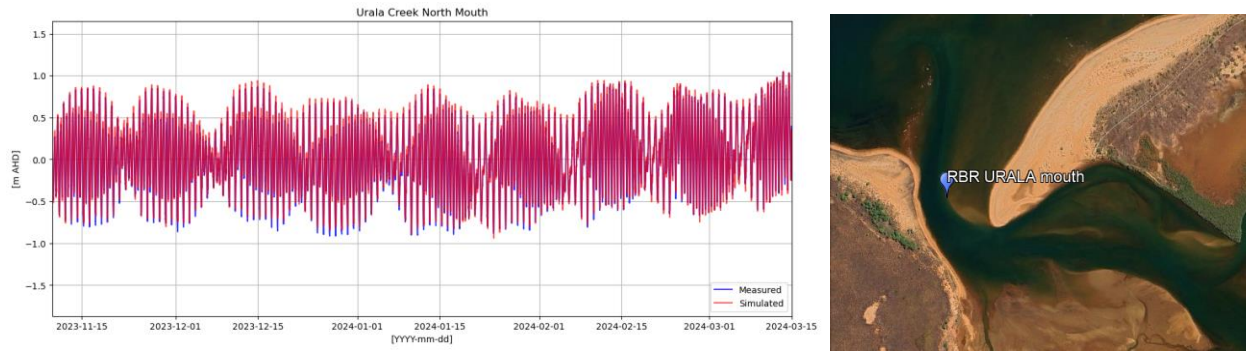


Figure 7 Measured and simulated water level at the Urala creek north mouth in left panel the location is highlighted in right. RBR is the water level measuring instrument.

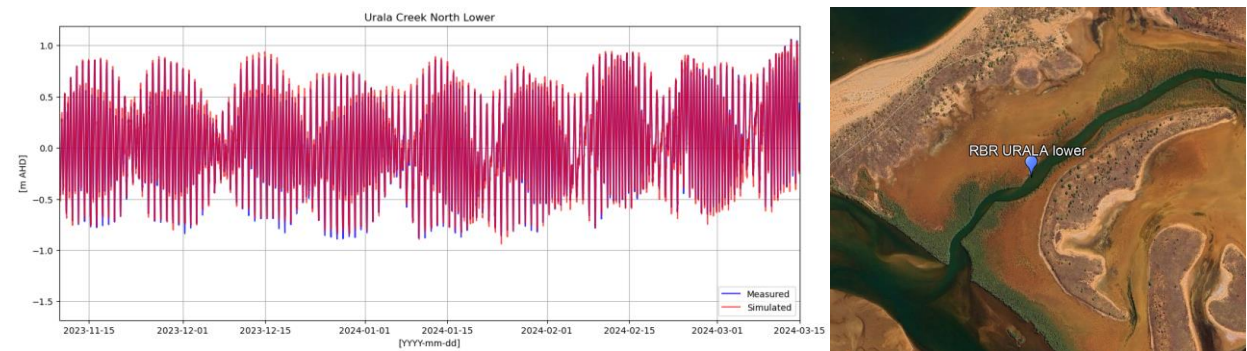


Figure 8 Measured and simulated water level at the Urala creek north lower in left panel the location is highlighted in right. RBR is the water level measuring instrument.

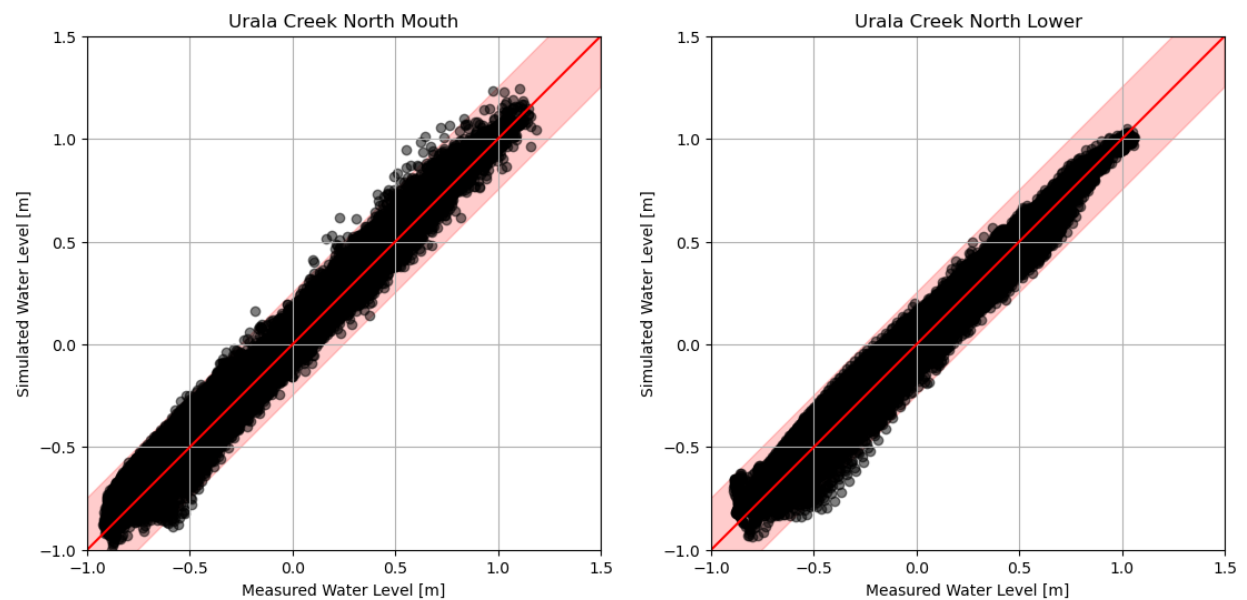


Figure 9 Scatter plot of measured vs simulated (1 to 1) water level comparison for both Urala creek north mouth and lower. The red shaded area indicates the 25% error bar.

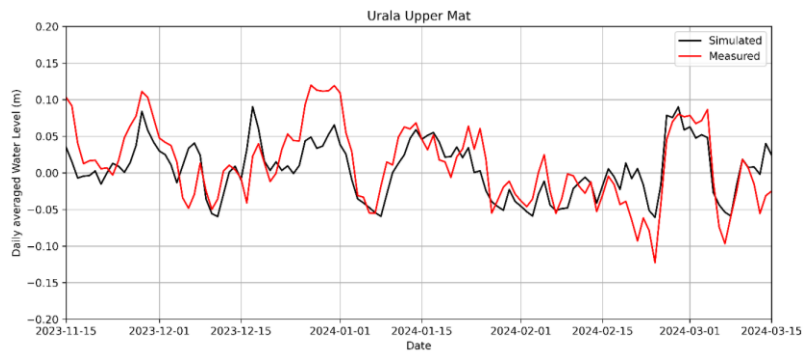


Figure 10 Daily averaged measured and simulated water level comparison at the Urala upper region.

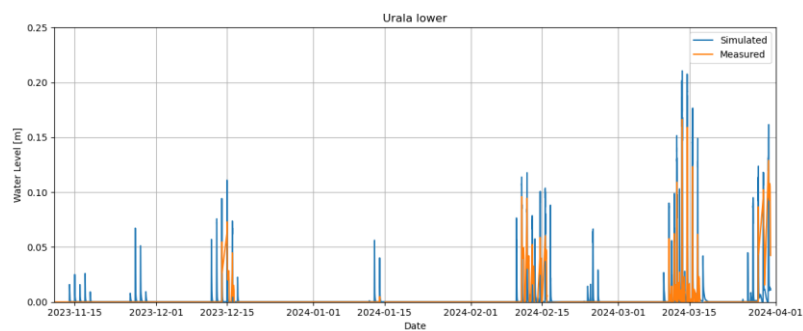


Figure 11 Measured vs simulated water level at the Urala lower mat.

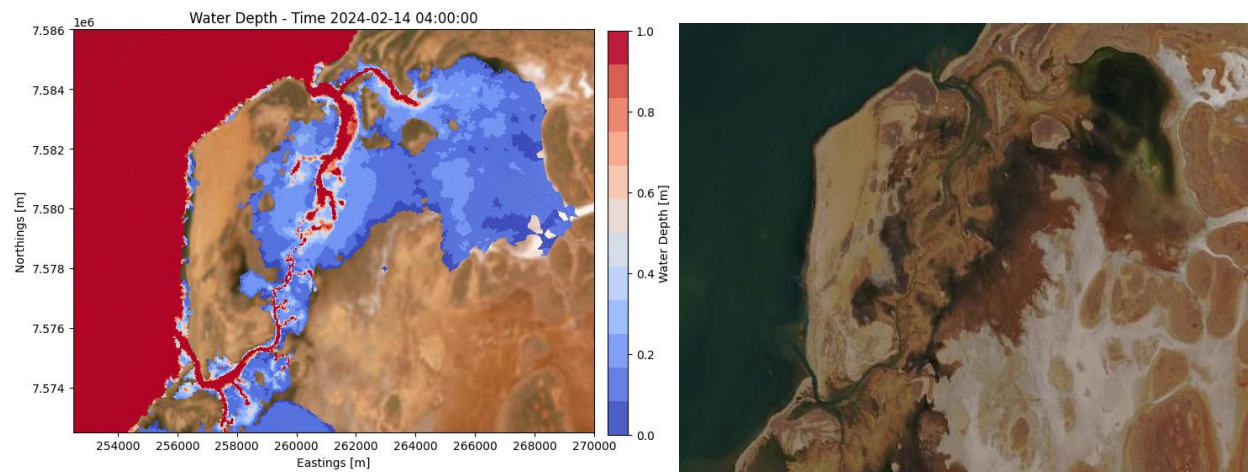


Figure 12 Simulated inundation extent during high tide on 14/02/2024 compared to the satellite observed extent on the same day.

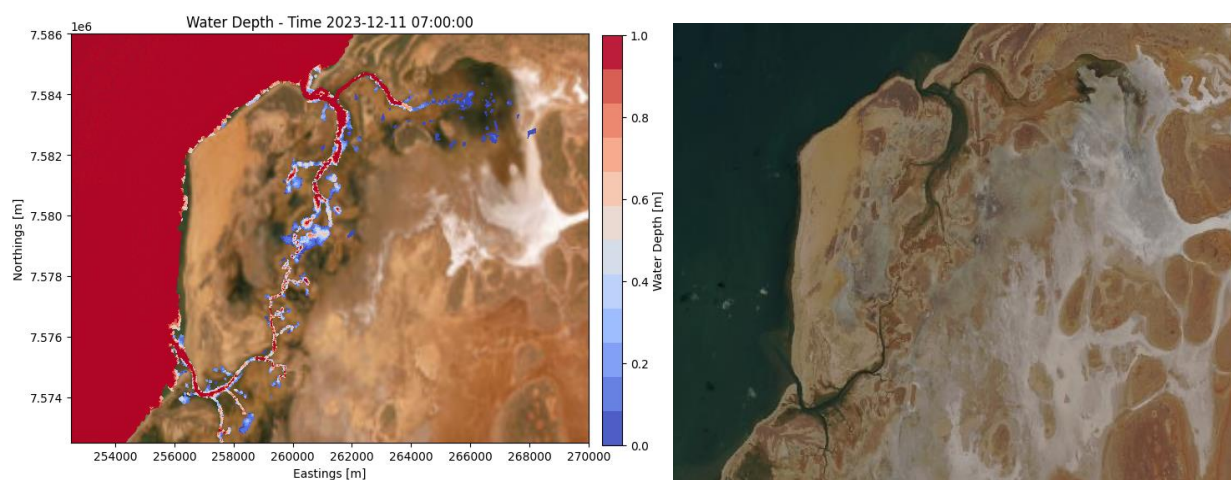


Figure 13 Simulated inundation extent during low tide on 11/12/2023 compared to the satellite-observed extent on the same day.

4. Habitat modelling

To predict habitat distribution under future SLR scenarios, we used a combination of hydrodynamic modelling and a machine learning approach (Random Forest algorithm). The hydrodynamic characteristics for different habitats were simulated using Delft3D Flexible Mesh (DFM), an open-source model widely used for simulating flow, sediment transport, and water quality in complex coastal environments (Kernkamp et al., 2011) (see Section 3 for model details and validation). The historical inundation regime derived from the hydrodynamic model was then used to train the Random Forest (RF) model (Breiman, 2001).

RF is a widely used supervised machine learning method for classification and regression (Oliveira et al., 2012). It has two key advantages that make it suitable for habitat mapping applications (Vrieling et al., 2018). First, as a non-parametric rule-based algorithm, it makes no prior assumptions about the relationship between predictor variables and the response, which allows for better performance in complex systems compared to parametric methods (Breiman, 2001). Second, RF can effectively handle non-linearity and interactions among predictors, outperforming generalized linear and additive models (Oliveira et al., 2012).

The RF model was trained using a stratified approach, ensuring equal representation of all habitat classes to avoid class imbalance in predictions. The training dataset was curated by filtering hydrodynamic variables based on the 25th to 75th quantile of their distribution. The selected predictors included key inundation-related parameters such as:

- inundation frequency,
- percentage of submergence,
- number of dry days,
- number of consecutive dry days,
- mean dry days,
- total number of wet/dry events (based on hourly data),

- depth-averaged mean and maximum currents, and
- mean and maximum water depths.

These hydrodynamic parameters were chosen to provide a comprehensive representation of habitat exposure to tidal and submersion dynamics, which are known to be critical determinants for predicting habitat migration patterns under changing environmental conditions.

Figure 14 shows the flowchart of RF approach illustrating the decision tree process and prediction, which is derived from the average of multiple decision trees. The model was trained by selecting an optimal range of hyperparameters, including the number of trees, depth of trees, and minimum samples per split, to achieve the best predictive performance. A systematic tuning process was conducted, evaluating multiple combinations of these parameters to balance accuracy and generalizability. The final model configuration was determined based on the best fit obtained after 1,000 training epochs, ensuring robust classification of habitat types. This iterative training process helped refine the model's ability to capture complex relationships between hydrodynamic predictors and habitat distribution, minimising overfitting while maximising predictive reliability. The hydrodynamic model runs (including all scenarios) and RF model training were conducted as part of the WAMSI Mardie Offset Research Project B using the Setonix supercomputer within the Pawsey Supercomputing Centre.

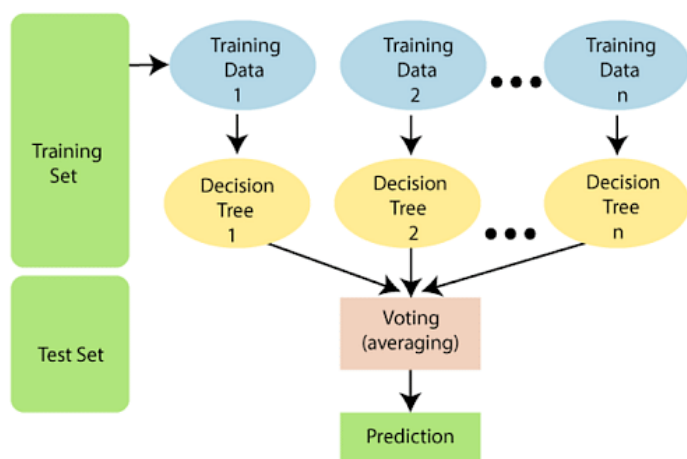


Figure 14 Flowchart of the Random Forest (RF) approach illustrating the decision tree process and prediction, which is derived from the average of multiple decision trees. Image Source: Simplilearn.com.

The model was applied to habitat mapping data for the years 2021 and 2023, which both had different spatial extents of algal mat and salt flat habitats. 80% of the data was reserved for training, and 20% of the data was reserved for validation.

4.1. Habitat prediction validation

Before proceeding with future habitat migration predictions, we sought to assess the model's predictive capability under current conditions. To achieve this, we tested the model using data from the representative year 2023, as well as an additional dataset from 2021 to evaluate its performance under different hydrodynamic conditions. Figure 15 shows box-whisker plots

illustrating the distribution of the various environmental factors for key habitats used to train the Random Forest (RF) model.

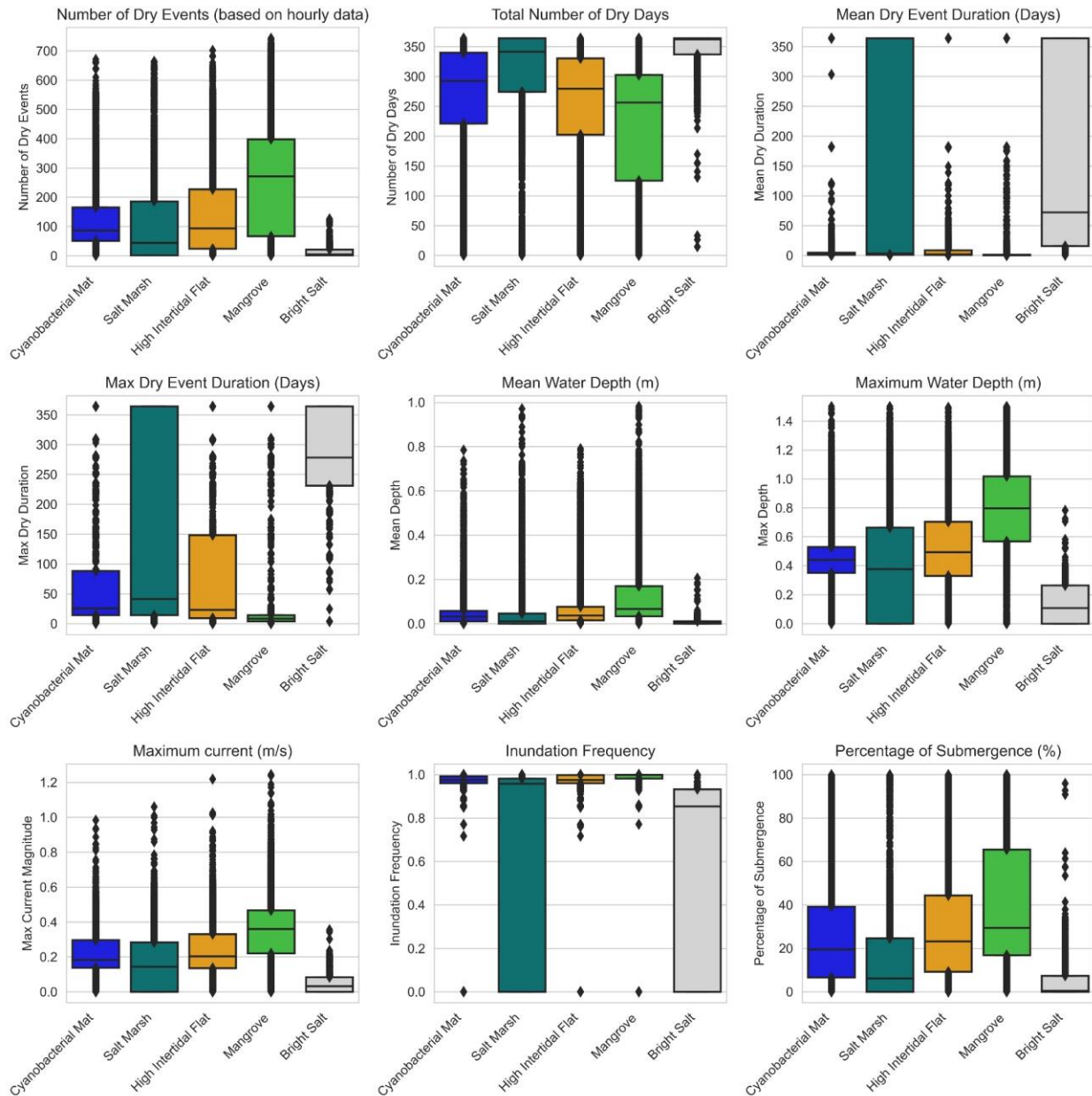


Figure 15 Environmental factors and their distributions for different habitats based on hydrodynamic conditions in 2023.

While the extracted data could potentially be used to develop a threshold-based model for predictions, the overlap of conditions across several variables made our initial attempts at this challenging with relatively weak results. In contrast, the RF approach leverages a more sophisticated fitting technique that applies different weights (feature importance) to different habitats and environmental conditions.

For future predictions, as hydrodynamic conditions are expected to change with sea-level rise (SLR), it is essential to train the model with adjusted hydrodynamic data that reflects water level changes and their corresponding impacts on habitat distributions. To this end, we also utilised data from 2018, a neutral ENSO year, and simulated both a 10 cm increase and decrease in water levels to account for potential ENSO-positive and ENSO-negative scenarios. For the ENSO-related cases, due to a lack of mapped habitat data, we relied on the 2021 habitat distribution maps.

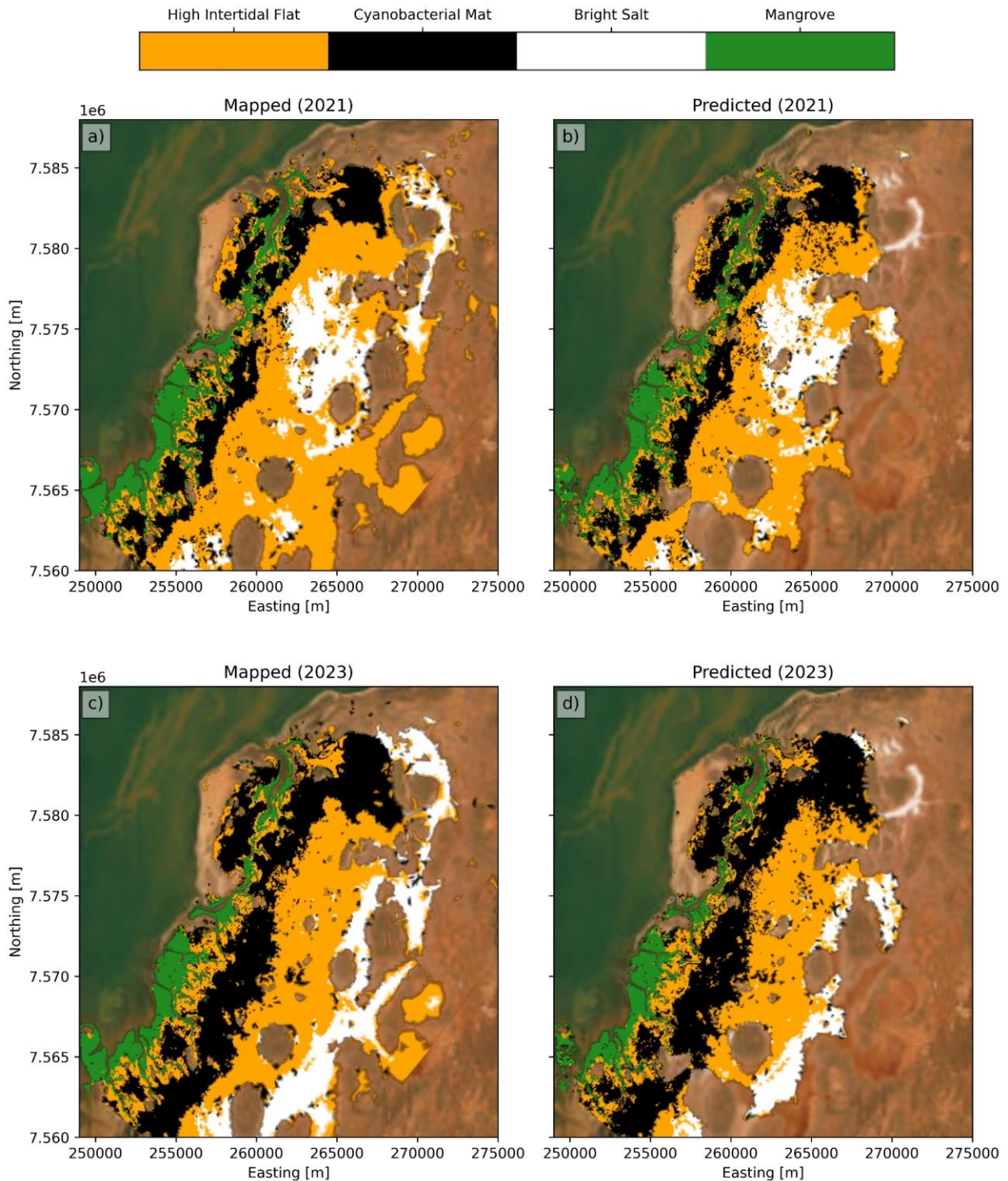


Figure 16 Mapped and RF predicted habitat classes for the year 2021 and 2023 hydrodynamic conditions.

Figure 16 presents the mapped and predicted habitats generated using the RF model for both 2021 and 2023. It is essential to note that, since the RF model was trained under hydrodynamic conditions, only cells that are inundated (i.e., cells that become wet) will have corresponding environmental factor data for prediction. Cells that do not get wet are also included in the prediction, as they were classified as “other classes” during training. The prediction also resulted in “other classes” because no environmental data is available for those areas. This limitation is one of the key reasons why rainfall was excluded from the analysis, as incorporating it produced hydrodynamic characteristics that were unrelated to the inundation region, leading to inaccurate habitat predictions. The inclusion of rainfall resulted in erroneous predictions in areas that were distant from any inundated zones, ultimately reducing the model's accuracy.

While the comparison between mapped and predicted habitats demonstrates reasonable accuracy, it also reveals some misclassification in specific areas, especially in regions where habitats are predicted in non-inundated zones. These misclassifications are expected, given the limitations of the model and its reliance on hydrodynamic data. The model successfully predicts habitats only for the inundated areas, so it does not generate predictions for non-inundated zones.

To further evaluate the prediction performance, we present the confusion matrix of predicted vs. mapped habitat data in Figure 17. This matrix is based on 20% of the dataset, which is reserved for model testing, with the remaining 80% used for training. The confusion matrix shows good agreement between the mapped and predicted data for all habitat classes, providing confidence in the model's ability to accurately predict habitat distributions. The relatively low misclassification rate in the confusion matrix suggests that the RF model effectively identifies key habitat features within the inundated regions.

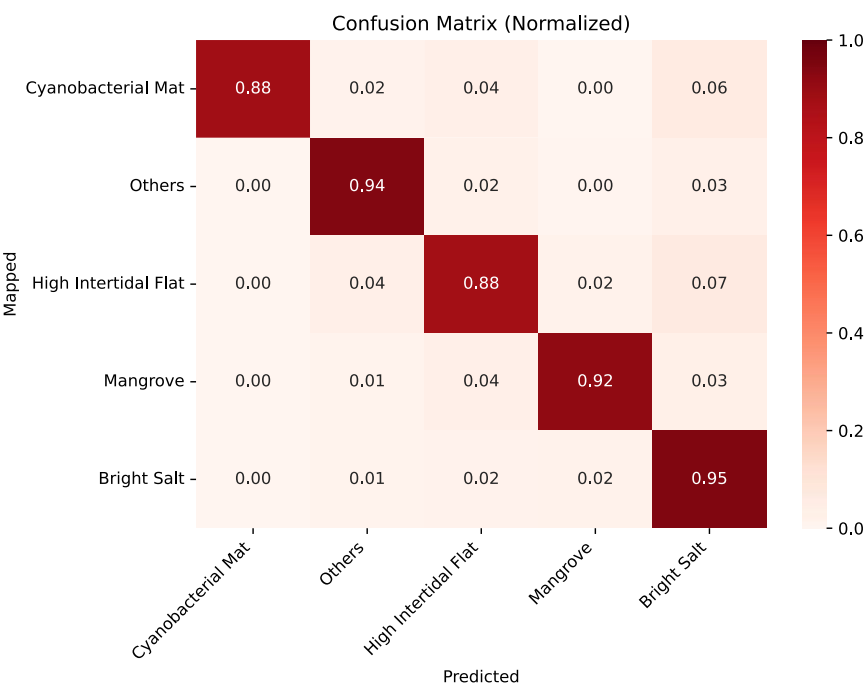


Figure 17 Confusion matrix (normalized) showing the accuracy of predictions.

To evaluate the relative influence of various hydrodynamic parameters in predicting habitat distribution, we extracted the feature importance scores derived from the optimised RF model. These scores offer insight into the extent to which each parameter contributes to the model's predictive performance. As illustrated in Figure 18, water depth (both maximum and mean) and depth-averaged current velocity (both maximum and mean) emerged as the most influential factors in the prediction process. These variables likely reflect the physical conditions most directly affecting habitat suitability. Following these, the mean number of dry days and the frequency of inundation were also found to be important contributors, suggesting that the duration and frequency of exposure to water play a critical role in shaping the spatial distribution of intertidal habitats.

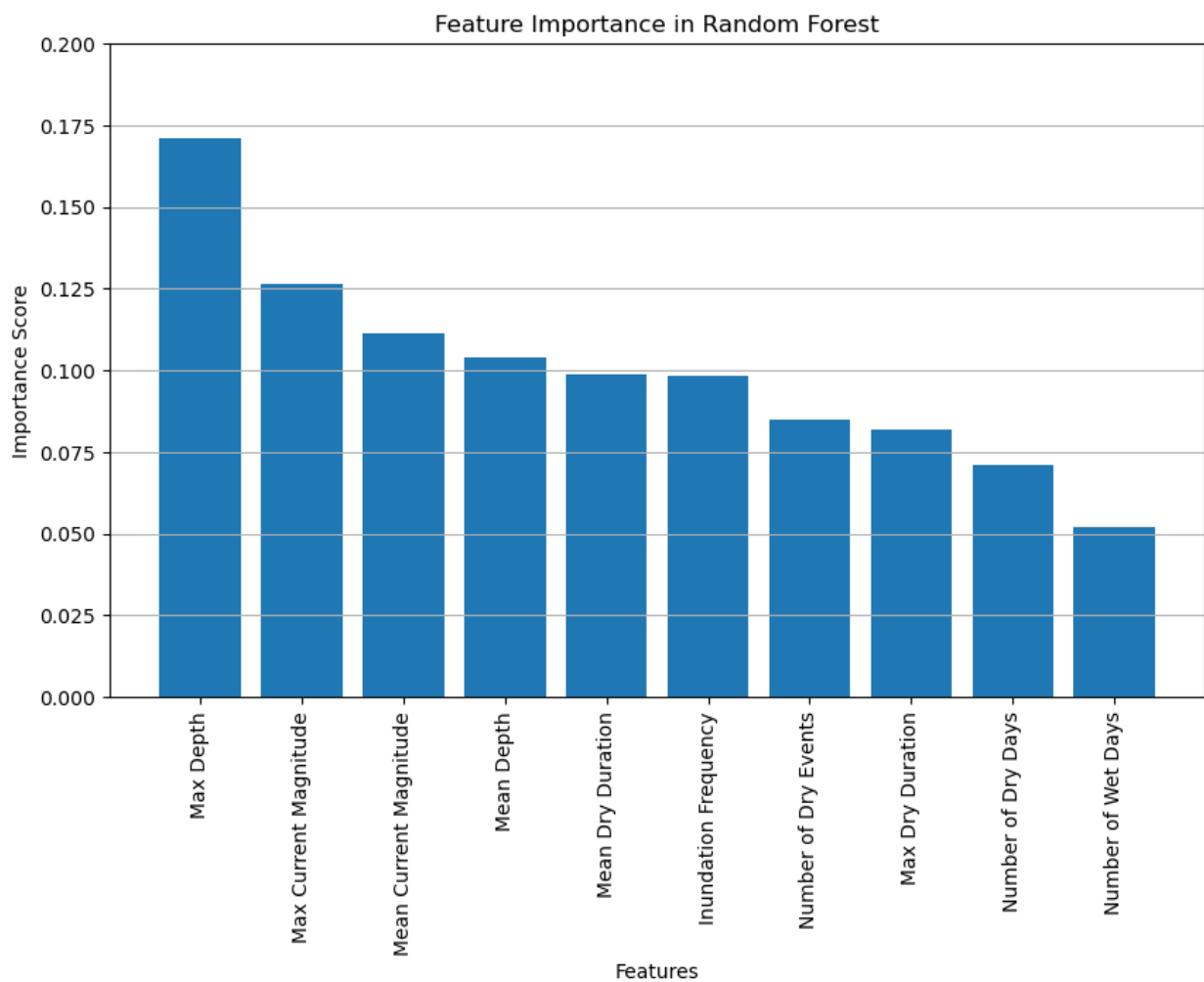


Figure 18 Importance score as optimized in RF approach by the model for various hydrodynamic parameters.

5. Scenario rationale

To predict the influence of SLR on intertidal habitats, we considered three scenarios based on different Shared Socioeconomic Pathways (SSPs), namely SSP245, SSP370, and SSP585, which

represent potential greenhouse gas emission trajectories. These correspond to moderate, high, and very high greenhouse gas emissions, respectively. These scenarios present a range of potential future sea level changes under various climate policy and economic development scenarios.

For this study, the projected SLR values were derived from the median average of these three scenarios—the adopted SLR projections for the years 2050, 2075, and 2100 as shown in Figure 19. With 2023 serving as the baseline scenario for comparison, the average annual increase rates are approximately 8.3 mm/year from 2023 to 2050, 8.6 mm/year from 2050 to 2075, and 11.8 mm/year from 2075 to 2100. The increasing rate of sea level rise highlights the accelerating impact of climate change, particularly in the latter half of the 21st century (Hague et al., 2020). This aligns with recent studies indicating that higher emissions scenarios (e.g., SSP585) could lead to a significant rise in global mean sea levels, potentially exceeding 1 meter by 2100 if ice sheet instability is triggered (Allen et al., 2018; Calvin et al., 2023).

To assess the potential impacts of the development scenario, we ran all simulations with and without the salt-pond development boundary, and the results are summarized in Table 4. In Delft3D FM, the development scenario is modelled by incorporating thin dam walls at the development boundary. Thin dams are infinitely thin structures defined at velocity points, preventing flow exchange between adjacent computational cells without altering the total wet surface or model volume.

In this study, the thin dam represents the presence of a development-related salt pond, which has sub-grid dimensions but is significant enough to influence local flow patterns. The thin dam is assumed to have an infinite elevation within the model, meaning that no water will ever overtop it, effectively acting as an impermeable barrier.

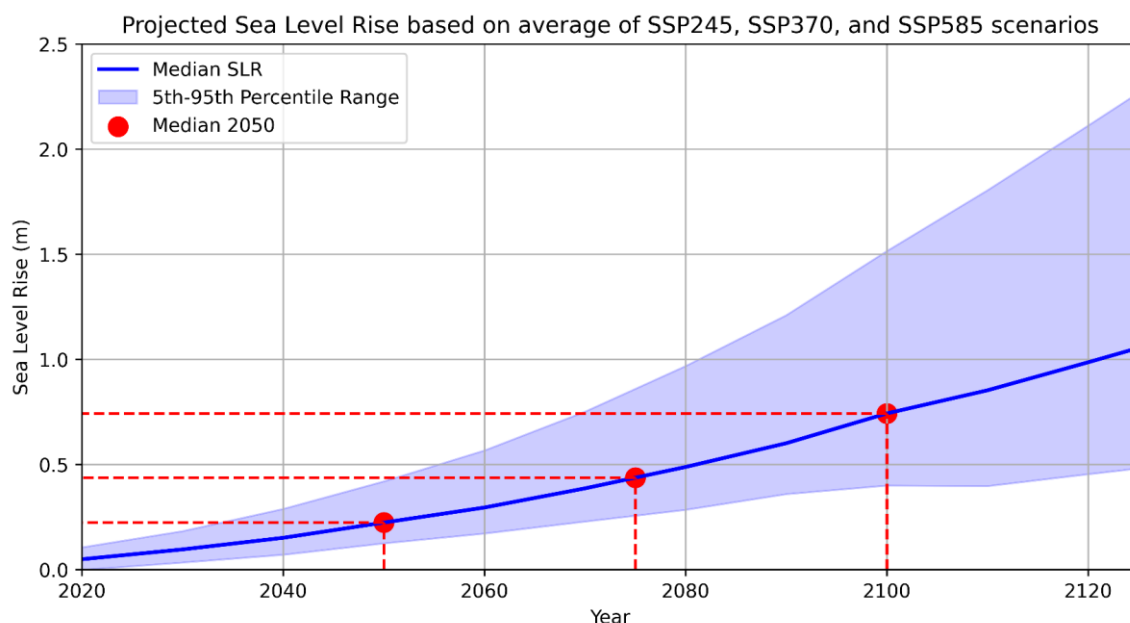


Figure 19 Projected sea level rise based on average of SSP245, SSP370 and SSP585. The red dots and dotted lines indicate the adopted SLR scenarios for this study. SSP - Shared Socioeconomic Pathways.

Table 3 Scenarios used for the habitat predictions.

Scenario	Representative year (no SLR)	SLR 2050	SLR 2075	SLR 2100
Baseline (without development)	2023	+ 0.224 m	+ 0.438	+0.734
Development	2023	+ 0.224 m	+ 0.438	+0.734

6. Future predictions

6.1. Baseline (Without Development)

Figure 20 shows the predicted habitat distributions under different SLR scenarios for the baseline (without development) condition. Compared to the no SLR case, all the SLR scenarios showed habitat shifts, indicating that the habitats tend to follow a set of certain suitable environmental conditions, sometimes captured through a Habitat Suitability Index (HSI).

As SLR increases, the extent of inundation expands further inland, which is expected. With the expanding extent of inundation, the habitats also tend to follow their suitable environmental conditions. Notably, cyanobacterial mats migrate inland, with high intertidal flats forming around them, e.g., Figure 20 b, c and d. A noticeable salt flat develops at the outer edge of the inundation zone, reflecting a zonation pattern consistent with what was observed in the field. This trend is evident for the 2050 and 2075 SLR scenarios, where the habitat distribution follows a clear spatial organization. However, with a further increase in SLR, for 2100, the zonation pattern becomes more mixed compared to the baseline condition (Figure 20a) For this case, severe loss of mangroves is also observed. It is important to highlight that areas with maximum water depth greater than 1m exhibit conditions similar to the creeks, which are set to be classified as “other classes”. The empty regions in the high intertidal areas (Figure 20d) are possible due to this issue. The hydrodynamic conditions/parameters used to predict the habitat predictions are presented in [Appendix A](#).

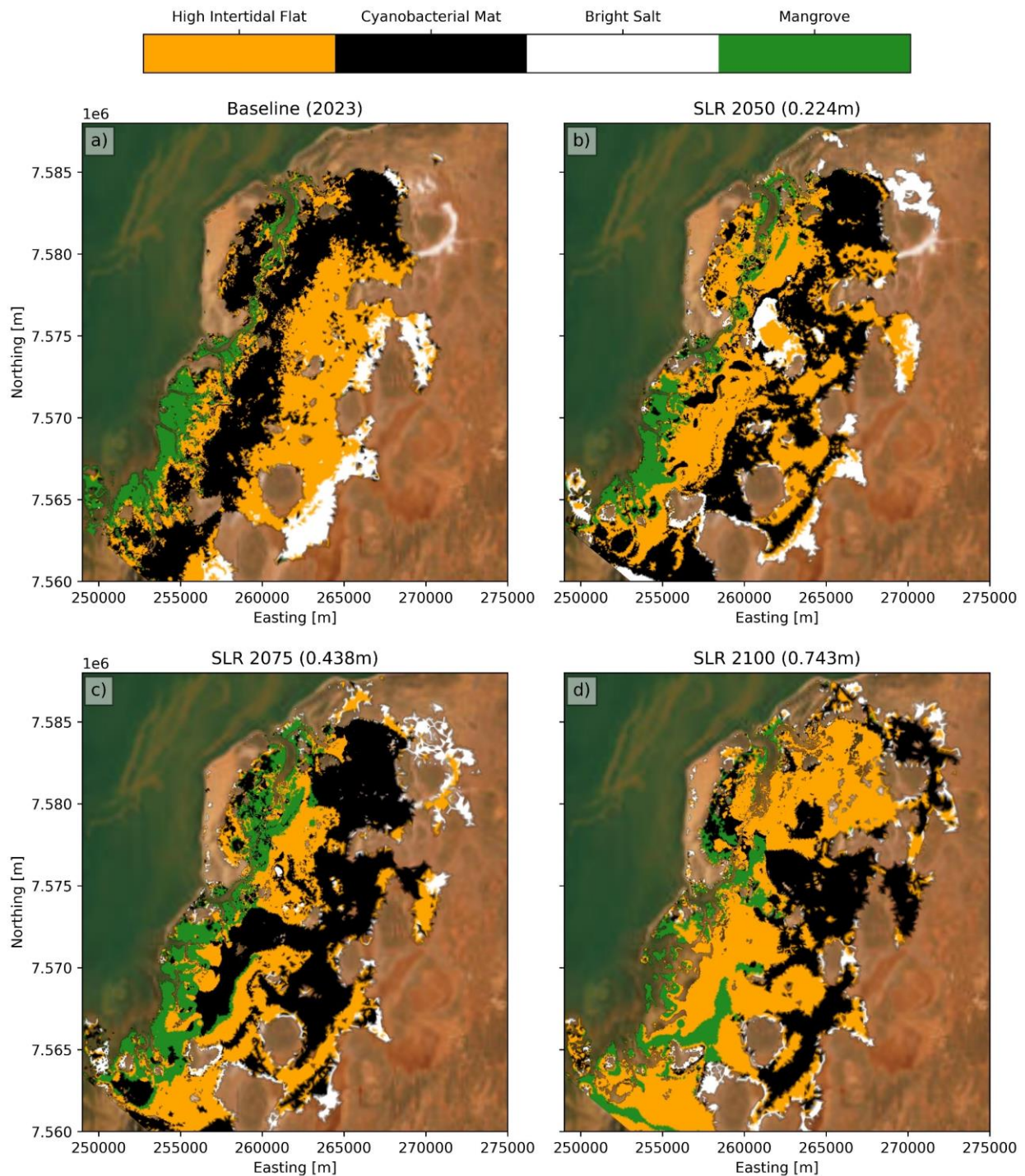


Figure 20 Predicted habitat distribution under different SLR scenarios (see Table 3) for the baseline condition.

6.2. With Development

Figure 21 shows the predicted habitat classes and their distribution under different SLR scenarios for the development condition. Due to the presence of the physical barrier, no habitats are predicted within the development boundary envelope (Figure 21). Outside the development boundary, a similar pattern is observed as in the scenarios without the development barrier, with marginal differences.

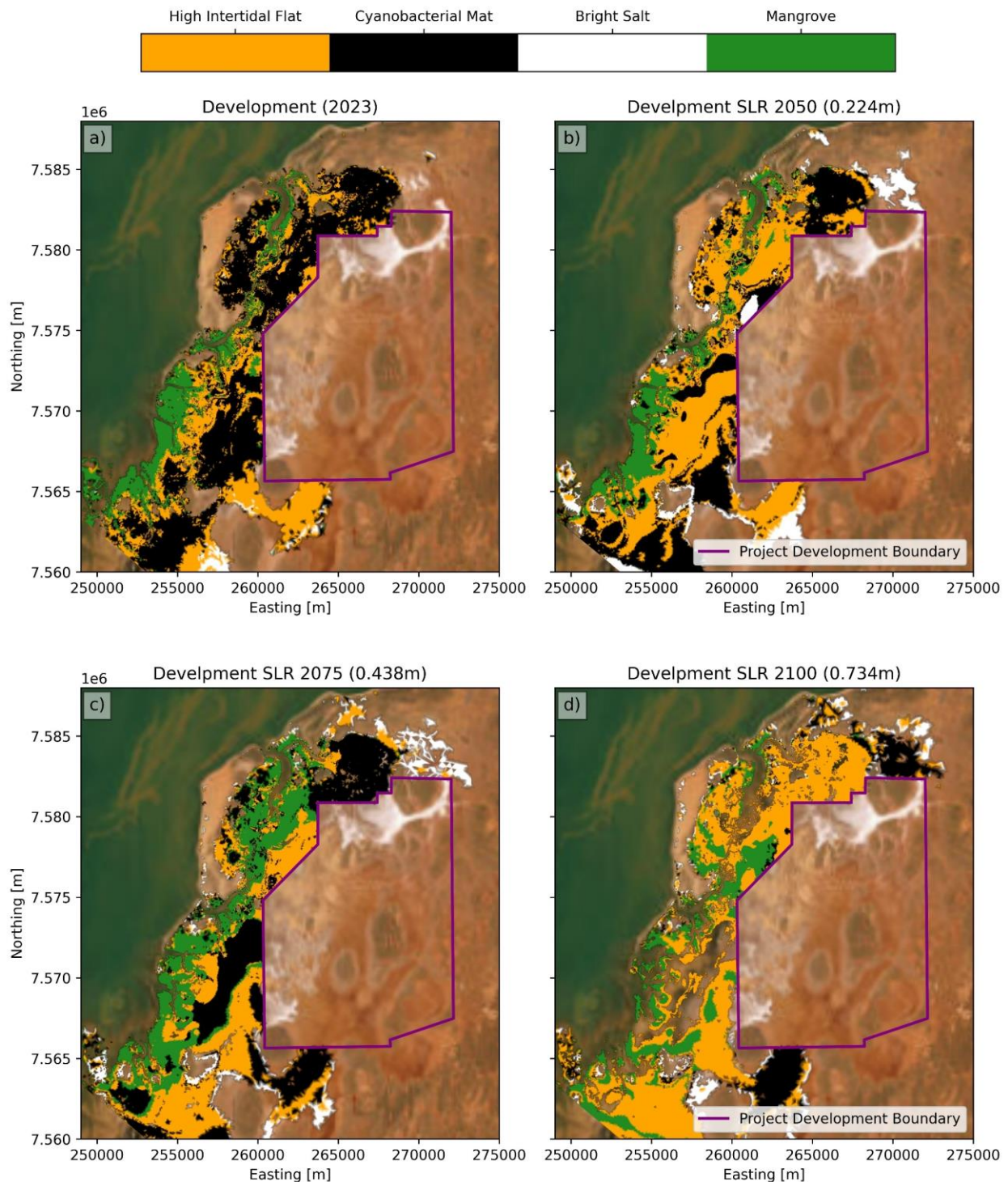


Figure 21 Predicted habitat classes and their spatial distributions under different SLR scenarios (see Table 3) for the development condition.

6.3. Habitat migration

To better assess the influence of SLR scenarios, we compared the habitat predictions for different classes separately. Figure 22 shows the mangrove distribution across various SLR scenarios. To illustrate the mangrove gain and loss resulting from SLR, we mapped the mangrove gain and loss, as well as locations where no change was observed with respect to the baseline scenario (Figure

22). As expected, the mangrove migration appears to be minimal with no significant changes for the SLR 2050 scenario compared to the baseline case, particularly in the highlighted area (Figure 22a). A noticeable portion of mangroves was lost in the southwest region of the domain (Figure 22b). However, with increasing SLR, we noticed a slight expansion of the mangrove footprint (Figure 22c). By SLR 2100, a significant reduction in mangrove cover is predicted. Figure 22 also predicted that mangroves migrate inland, particularly near the southern creek, whereas they showed less movement near Urala Creek North.

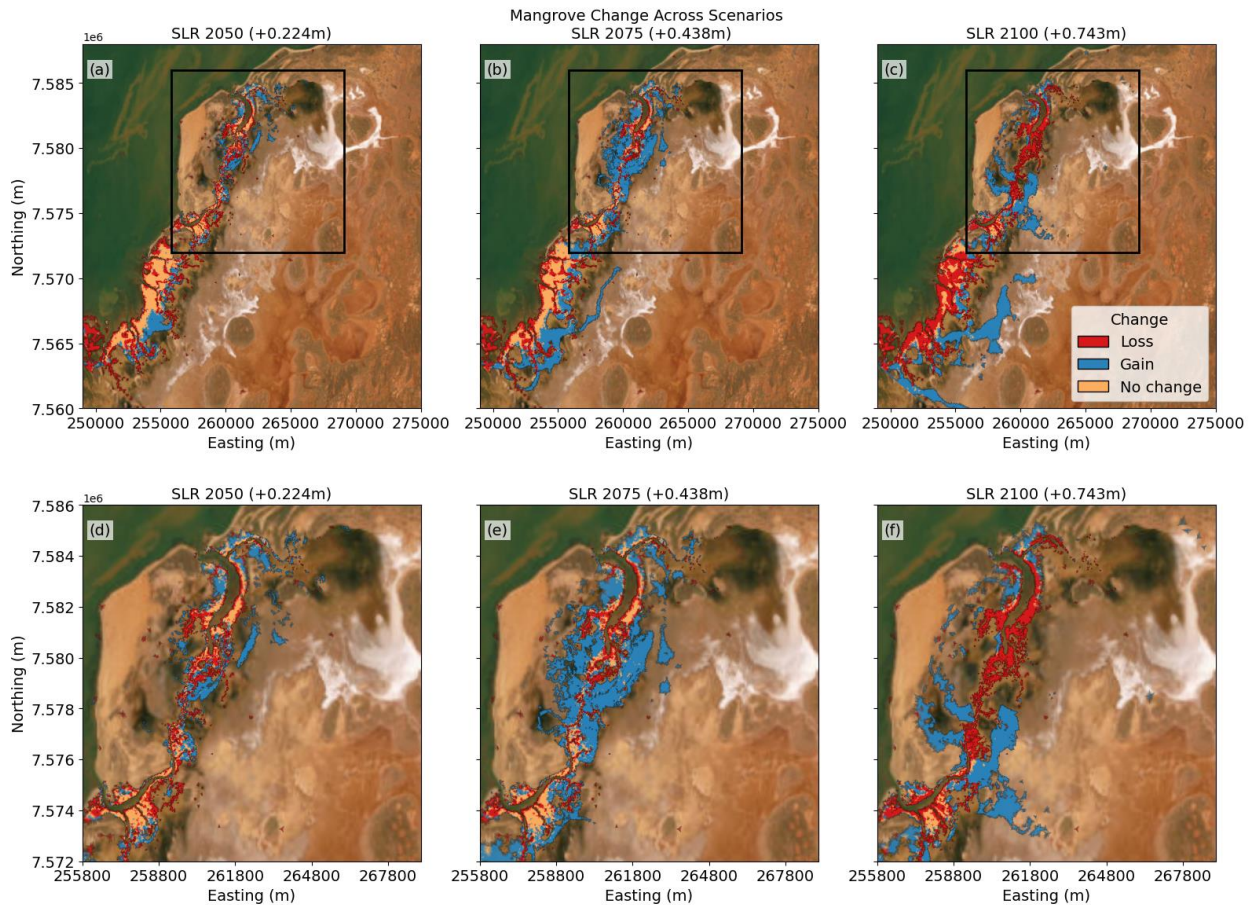


Figure 22 Mangrove distribution represented by gain and loss across different SLR scenarios compared to baseline(no SLR) without the development boundary envelope. The top panels show the overall domain, while the bottom panels show the zoomed-in view of the inset box shown in the top panels.

Cyanobacterial mats are complex, multi-layered microbial communities dominated by filamentous cyanobacteria that thrive in intertidal and supratidal environments, particularly in arid and semi-arid coastal systems. These mats play a critical ecological role by stabilising sediments, contributing to primary production, and influencing nutrient cycling within the ecosystem (Rybak et al., 2024). Their resilience to extreme environmental conditions—such as high salinity, desiccation, and intense solar radiation—makes them key indicators of environmental change (Demoulin et al., 2019). Cyanobacterial mats often form dense, photosynthetically active layers on mudflats and salt flats, where periodic inundation and exposure regulate their growth, activity, and dormancy phases (Demoulin et al., 2019; Rybak et al., 2024). These physiological adaptations allow them to persist and respond dynamically to hydrodynamic forces, and are also expected to adapt due to changes in sea level.

Similar to Figure 22, Figure 23 shows the distribution of cyanobacterial mats based on different SLR scenarios compared to the baseline scenario. Unlike mangroves, cyanobacterial mats exhibit more pronounced variations in spatial extent across different sea level rise (SLR) scenarios. The observed trend suggests that as sea level rises, these mats tend to migrate further inland, primarily driven by changes in inundation dynamics such as submergence duration and water depth. This inland shift is accompanied by a slight increase in their overall footprint up to the year 2075, indicating potentially more favourable hydrodynamic conditions for mat proliferation during this period. However, by 2100, the extent of cyanobacterial mats drops significantly, suggesting that increasing water depth may become a limiting factor for their persistence, likely due to reduced exposure times and light availability—both critical for their growth.

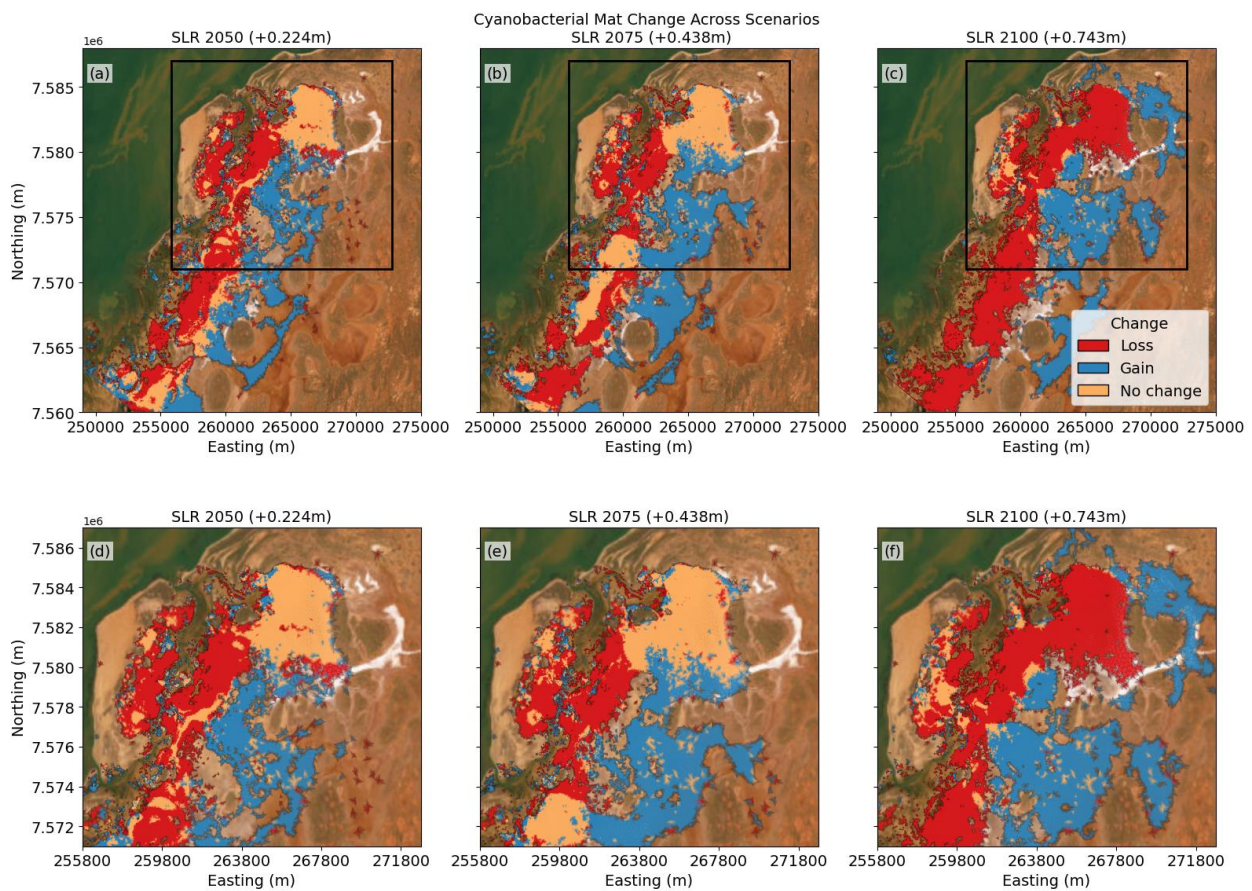


Figure 23 Cyanobacterial mat (algal mat) distribution represented by gain and loss across different SLR scenarios compared to baseline(no SLR) without the development boundary envelope. Top panels shows the overall domain while the bottom panels show the zoomed-in view of the inset box show in the top panels..

A comparable pattern is observed for salt flats (Figure 24), where the areal extent initially increases under the 2050 scenario but then steadily declines in the 2075 and 2100 projections. This may be attributed to shifts in the frequency and duration of drying phases, which are essential for salt flat development.

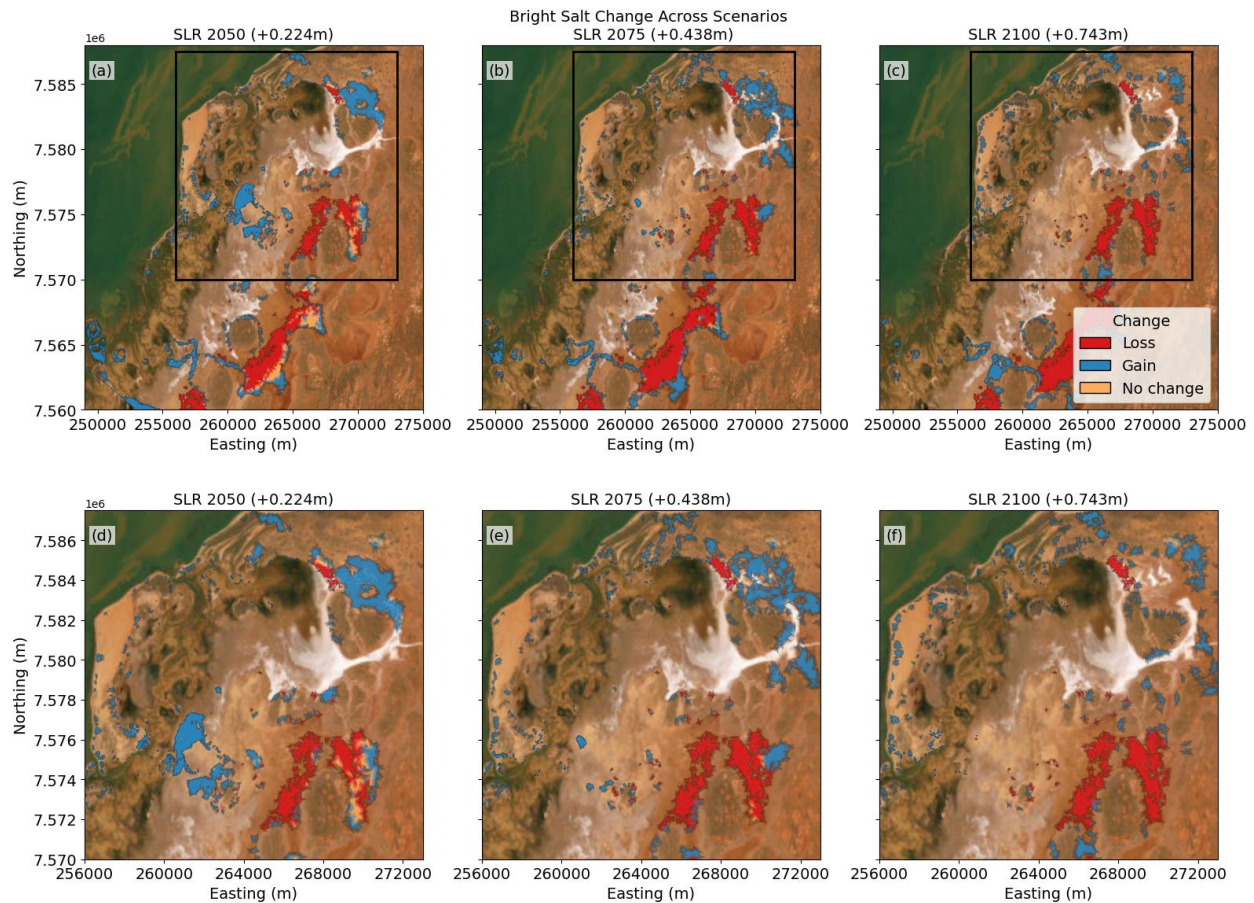


Figure 24 Salt flat distribution represented by gain and loss across different SLR scenarios compared to baseline(no SLR) without the development boundary envelope. Top panels shows the overall domain while the bottom panels show the zoomed-in view of the inset box show in the top panels..

It is important to note that the current predictions are solely based on hydrodynamic variables such as water depth, dry days, and submergence frequency. Key environmental parameters like salinity and temperature, which are known to have a significant influence on the distribution and health of all intertidal habitats, were not included in this analysis. These factors will be incorporated in future modelling efforts to provide a more comprehensive understanding of habitat shifts under changing climate and sea level conditions.

Consistent with the baseline predictions (which excluded the development boundary), we also analysed the individual habitat distributions under all sea level rise (SLR) scenarios while incorporating the development boundary, referred to here as the development scenarios. Figure 25 to Figure 27 shows these development scenarios for various habitats. In the simulations, the development boundary was represented as a thin impermeable structure (analogous to a thin dam wall), effectively blocking any hydrodynamic exchange between the area inside the boundary and the surrounding environment. As a result, no habitats were predicted within the development boundary under any of the scenarios.

Outside these boundaries, the predicted habitat distributions closely resembled those from the baseline scenarios, with only minor and negligible differences observed. This indicates that the presence of the development boundary had a localised impact, primarily restricting habitat formation within the enclosed area while exerting minimal influence on habitat patterns beyond

it. To further understand the potential impacts on the footprints, we calculated the total area for each habitat for all the scenarios.

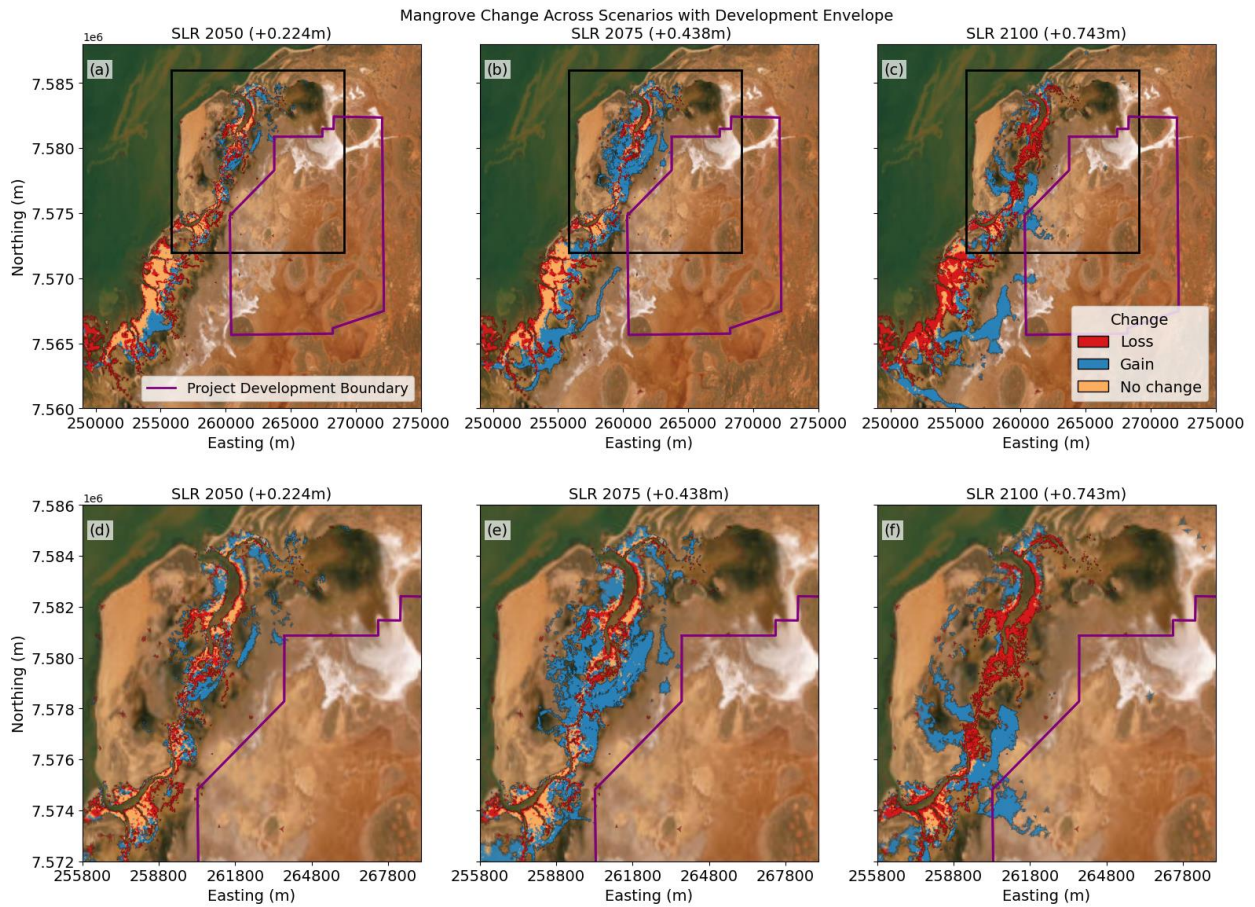


Figure 25 Mangrove distribution represented by gain and loss across different SLR scenarios compared to the development baseline(no SLR), with the development boundary envelope highlighted in purple colour. The top panels show the overall domain, while the bottom panels show the zoomed-in view of the inset box shown in the top panels.

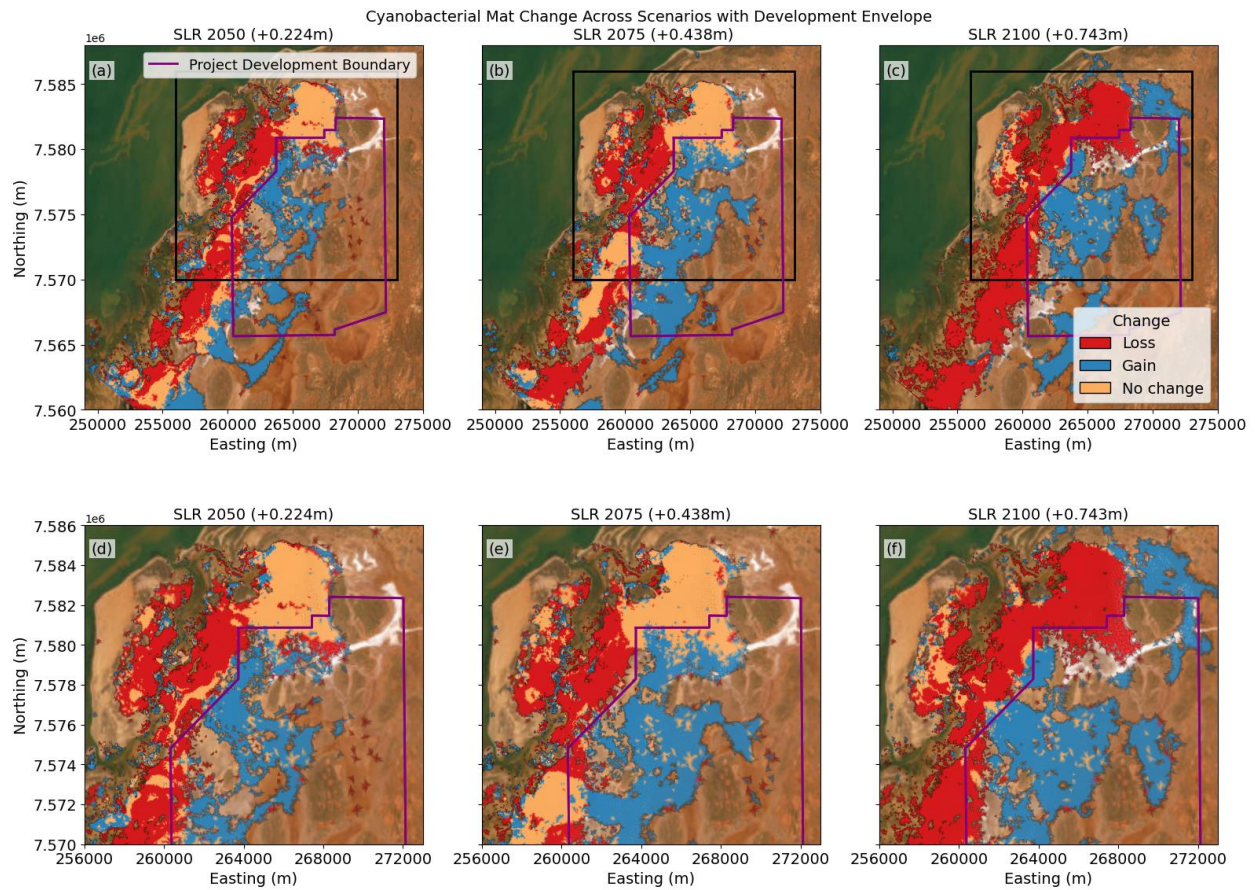


Figure 26 Cyanobacterial mat (Algal mat) distribution represented by gain and loss across different SLR scenarios compared to the development baseline(no SLR), with the development boundary envelope highlighted in purple colour. The top panels show the overall domain, while the bottom panels show the zoomed-in view of the inset box shown in the top panels.

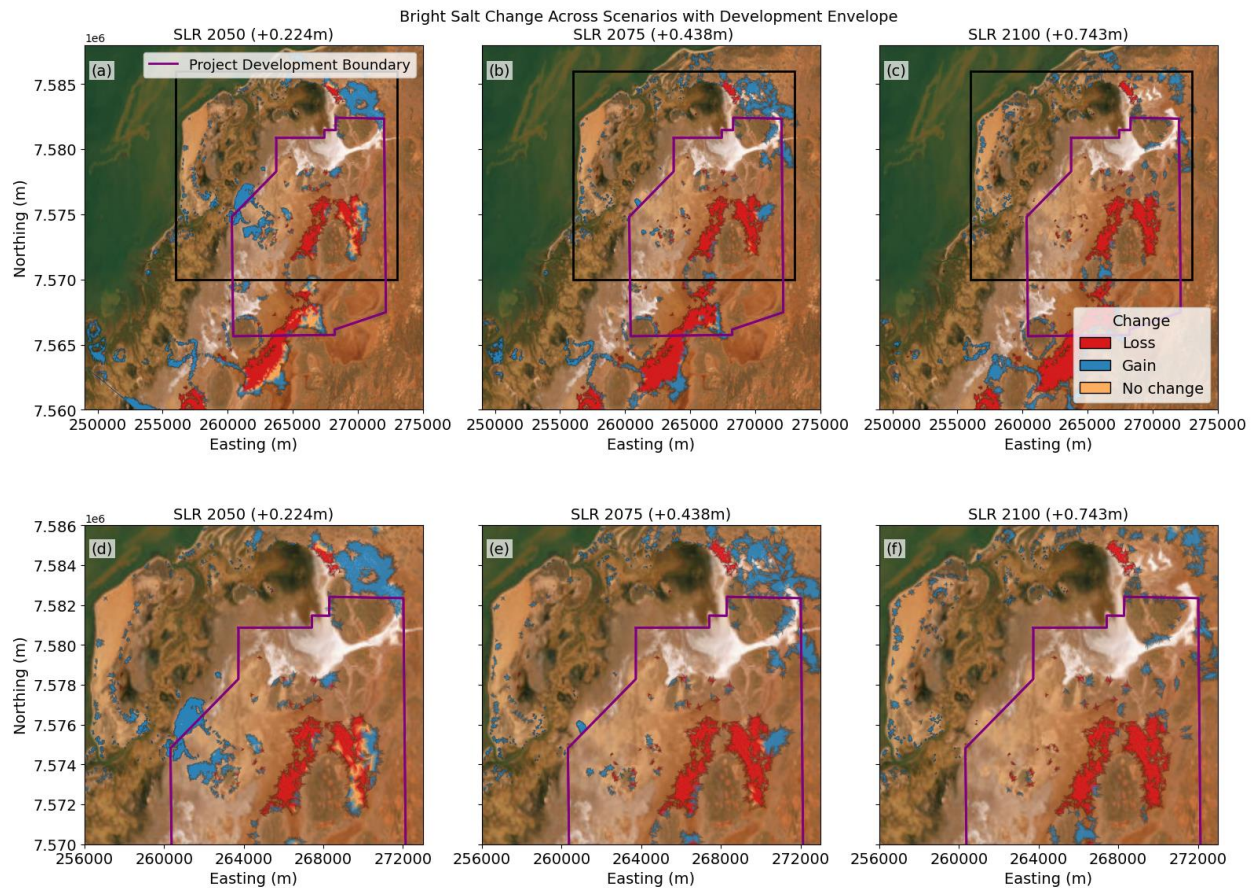


Figure 27 Salt flat distribution represented by gain and loss across different SLR scenarios compared to the development baseline(no SLR), with the development boundary envelope highlighted in purple colour. The top panels show the overall domain, while the bottom panels show the zoomed-in view of the inset box shown in the top panels.

Figure 28 shows the habitat area with and without the development boundary, and the same is tabulated in Table 4. The results indicate that the presence of the development boundary has a negligible impact on the spatial extent of mangroves, with minimal changes observed between scenarios with and without the boundary. In contrast, the cyanobacterial mat exhibits a noticeable reduction in habitat area due to the development, indicating that its distribution is more sensitive to hydrodynamic isolation. A similar trend is observed for the intertidal flat, which shows an overall decrease in footprint when the development boundary is included. It is also important to note that although the high intertidal flat was classified as a distinct habitat class, field observations revealed the presence of cyanobacterial mats beneath the muddy or sandy surface, suggesting these may represent dormant algal states. This highlights the dynamic nature of intertidal systems and indicates that some habitat transitions may not be fully captured through classification alone, especially under development-constrained hydrodynamic conditions.

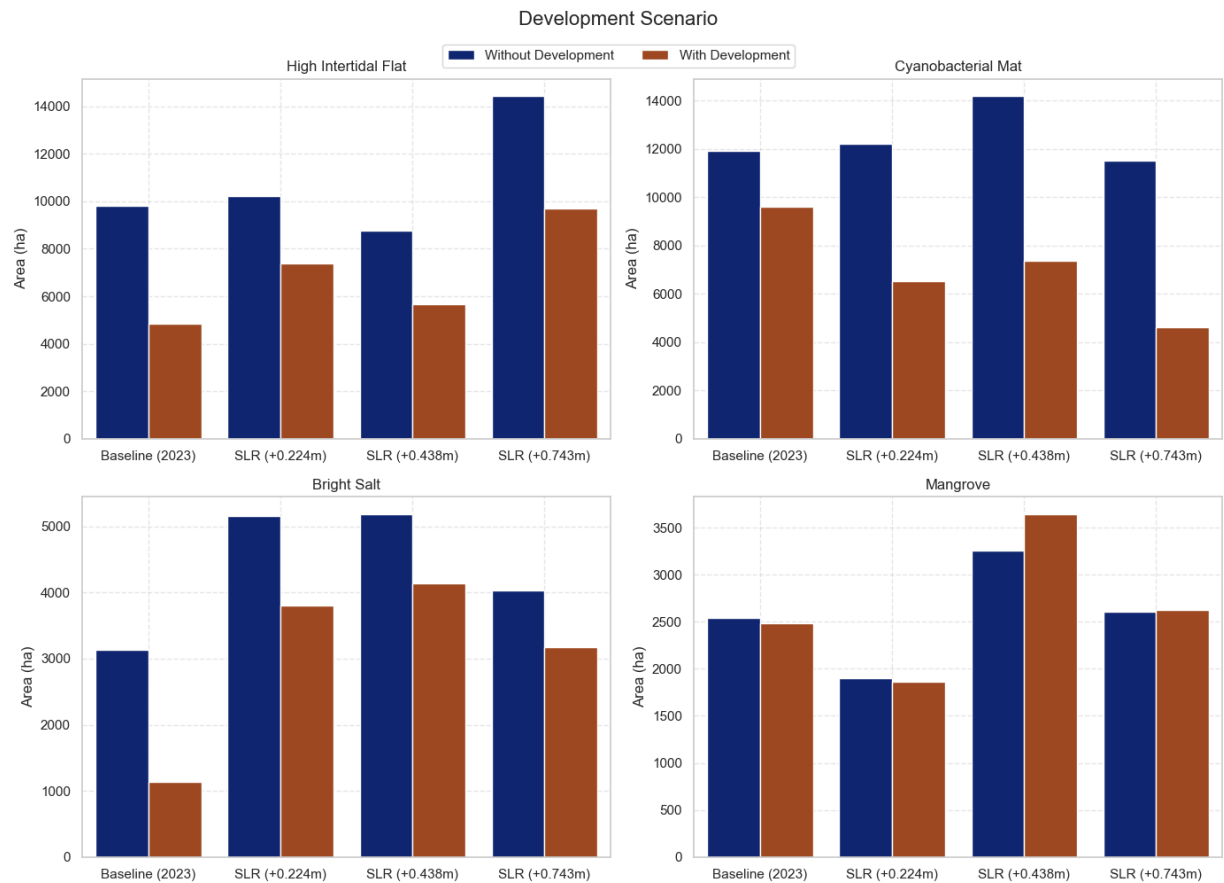


Figure 28 Predicted change in habitat area (in hectares, ha) without and with development footprint.

Table 4 Predicted change in habitat area (in hectares, ha) for different SLR scenarios without and with development footprint.

Habitats/Scenarios		Baseline (2023)	SLR 2050 (+0.224m)	SLR 2075 (+0.438m)	SLR 2100 (+0.743m)
High Intertidal Flat area (ha)	Without Development	9795	10204	8745	14437
	With Development	4840	7383	5662	9701
Cyanobacterial Mat area (ha)	Without Development	11930	12221	14204	11517
	With Development	9626	6540	7378	4616
Bright Salt area (ha)	Without Development	3133	5153	5184	4028
	With Development	1131	3797	4143	3179
Mangrove area (ha)	Without Development	2540	1901	3259	2608
	With Development	2486	1858	3644	2621

To gain a deeper understanding of habitat migration patterns, we separately extracted and combined the habitat distributions for all SLR scenarios, then excluded areas where these habitats were already present in the baseline (no SLR) scenario. This approach highlights potential migration zones for each habitat and helps identify key areas of interest. Figure 29 illustrates the migration patterns for individual habitats.

To assess potential habitat movement into the development boundary, we overlaid the development boundary on the migration plots shown in Figure 29. The results indicate that while only a small area of mangrove habitat migrates into the development zone, a significant portion of the cyanobacterial mat appears to migrate into this zone. Although these findings are based solely on habitat suitability based on hydrodynamic conditions, the extensive migration of the cyanobacterial mat into the development area suggests that this habitat may become increasingly suitable within the development zone in the future.

Furthermore, to identify potential future migration into previously unclassified areas, we applied a similar method as described above, but this time we combined all the habitat classes and excluded locations that were classified under any habitat category in the baseline scenario.

Figure 30 presents the future migration of combined habitats into areas that were not classified in the baseline. These areas include the expansion of salt flats and regions around the highland edges, where the inundation extent did not reach under baseline conditions.

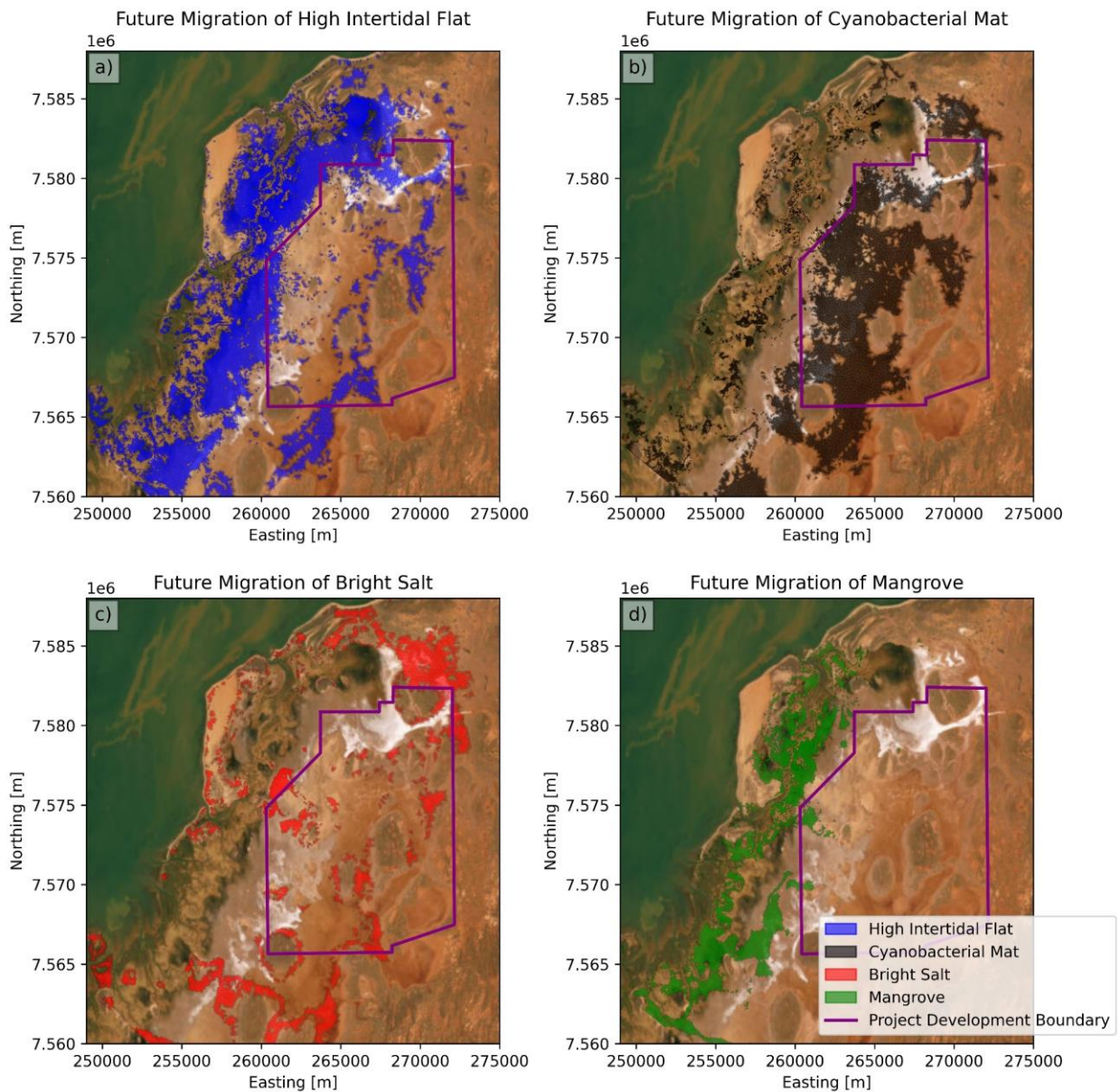


Figure 29 Future migration (all SLR scenarios combined) of habitats compared to the baseline scenario (no SLR).

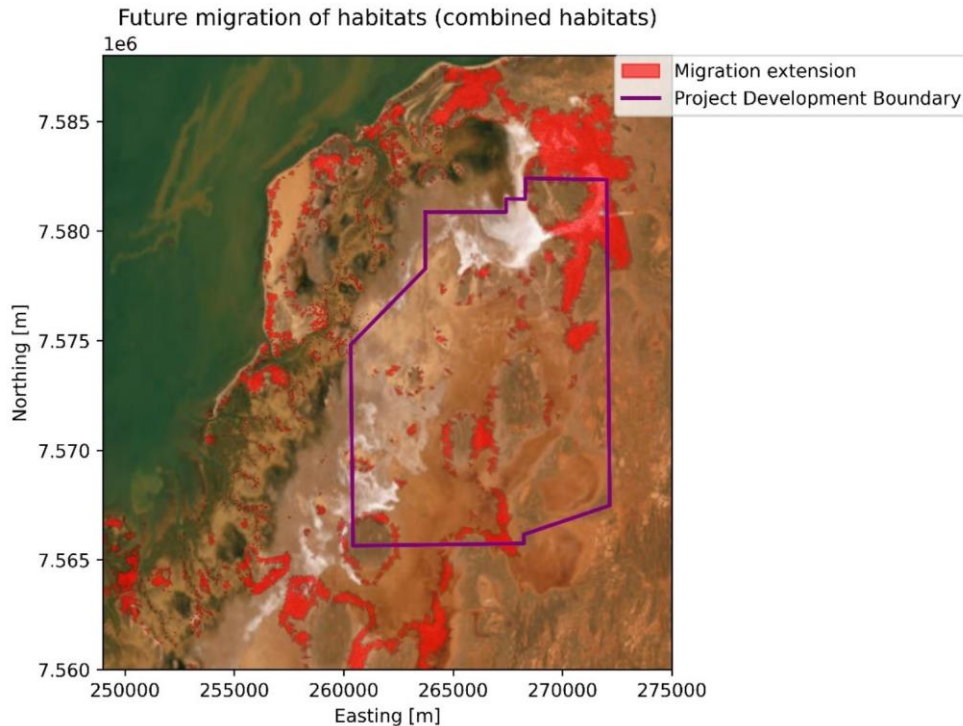


Figure 30 Predicted future habitat extent compared to the baseline distribution. The figure highlights newly identified habitat areas under future conditions that were not classified as any habitat type in the baseline (present-day) distribution, indicating potential habitat expansion or shifts due to changing environmental conditions.

6.4. El Niño–Southern Oscillation (ENSO) effects

The extent of inundation in Urala—and the Pilbara region more broadly—is driven by fluctuations in sea level that occur across a wide range of time scales. These fluctuations arise from multiple interacting components, including astronomical tides, atmospheric pressure-induced surges, seasonal and interannual water level variability, and long-term mean sea level trends (Lowe et al., 2021; Merrifield et al., 2021).

To better understand the relative contributions of these components to sea level variability, we employed a decomposition method consistent with the approach adopted by Lowe et al., (2021). Figure 31 shows the results of this decomposition, based on observed water level data from the Onslow tide gauge station available from the Western Australia Department of Transport (<https://catalogue.data.wa.gov.au/dataset/tide-stations>). To provide context for extreme events, we also overlaid the timing of Tropical Cyclones (TC) and Tropical Lows (TL) using historical cyclone records from the Australian Bureau of Meteorology (<http://www.bom.gov.au/cyclone/tropical-cyclone-knowledge-centre/history/past-tropical-cyclones/>) (Figure 31a). Water levels were adjusted from Lowest Astronomical Tide (LAT) to Australian Height Datum (AHD).

The interannual variability component, shown in Figure 31b reveals considerable fluctuations, ranging from approximately 0.1 m up to 0.4 m in magnitude. This variation is likely linked to broader climate oscillations, including El Niño–Southern Oscillation (ENSO) events. Additionally, the long-

term sea level rise trend observed at Onslow is estimated to be approximately 0.17 m over the recorded period.

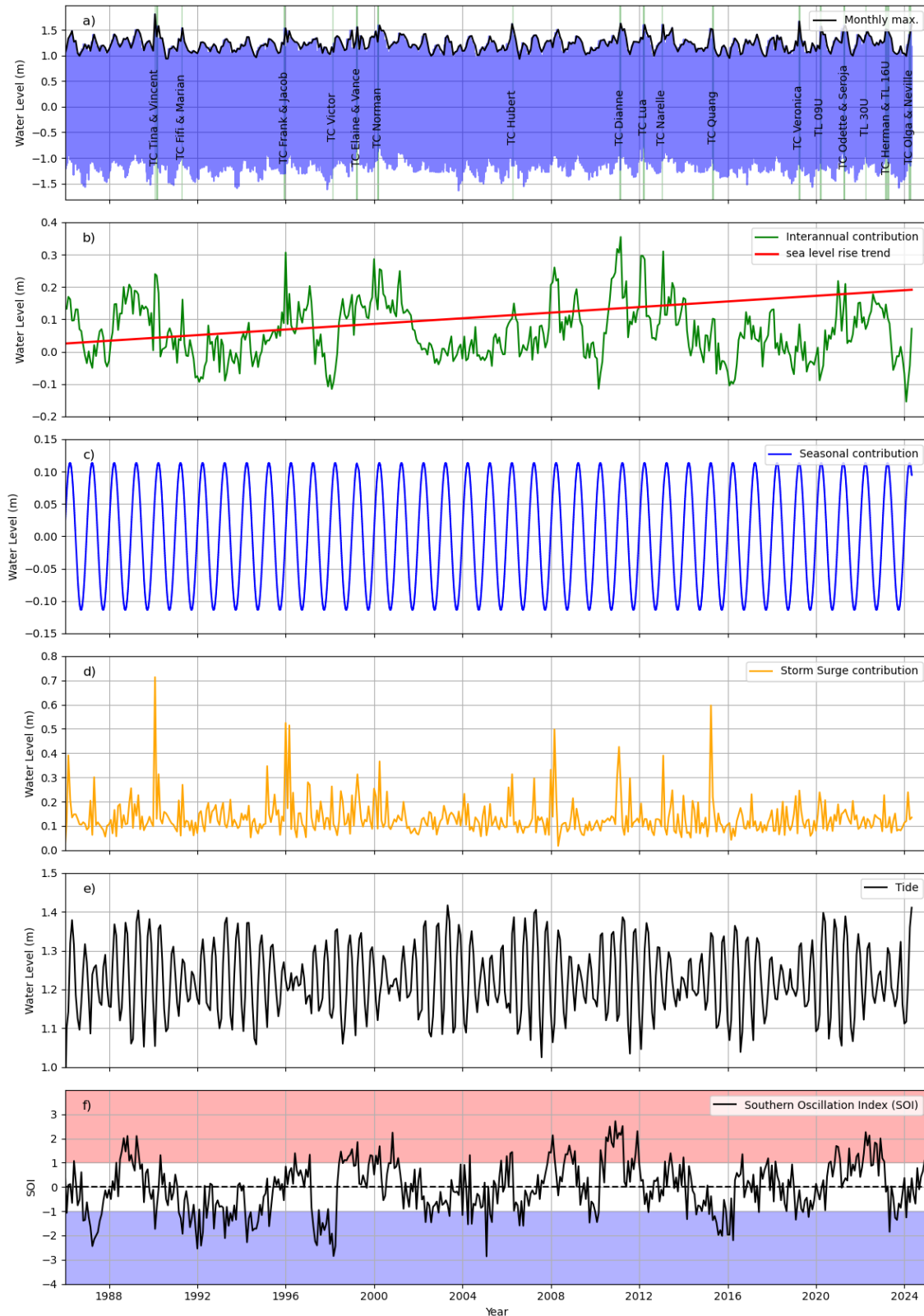


Figure 31 Decomposition of coastal sea level variability (Onslow, Western Australia), showing monthly maximum values for visual clarity. (a) Measured total water level (in blue) with monthly

maximum highlighted in black lines along with extreme conditions (Tropical Cyclones, TC and Tropical Low, TL) highlighted with green vertical lines, and the decomposed (b) interannual and mean sea level trend contributions, (c) seasonal contribution, (d) surge contribution, and (e) tide contribution. (f) Southern Oscillation Index (SOI), with highlights for values ± 1 indicating an increasing likelihood of El Niño and La Niña events.

Seasonal variability (Figure 31c) is also pronounced, with amplitude changes of up to 0.25 m. These variations are typically associated with seasonal shifts in ocean temperature, wind patterns, and regional atmospheric pressure systems. The atmospheric surge contributions Figure 31d represent short-term, meteorologically driven increases in sea level, particularly during cyclonic events. The monthly maximum tidal elevation variability range is between ~0.2 and 0.4 m (Figure 31e). These tides represent the largest and most predictable component of sea level variability, and they play a crucial role in defining the inundation envelope of coastal ecosystems. To better contextualise these patterns in relation to larger climatic drivers, Figure 31f shows the Southern Oscillation Index (SOI), a measure derived from the pressure difference between Tahiti and Darwin. The SOI is a key indicator of the phase and intensity of ENSO events, with sustained positive values indicating La Niña conditions and sustained negative values suggesting El Niño conditions (www.bom.gov.au/climate/enso/soi/). These climate states are known to significantly influence regional sea levels and storm activity in northwestern Australia.

It is important to highlight that each component contributing to coastal sea level variability—such as tides, atmospheric surges, seasonal cycles, and interannual fluctuations—has its own distinct phase and amplitude. When the phases of these components align constructively, their amplitudes can reinforce each other, resulting in extreme water level events. Such extreme events—whether wet or dry—can impose significant stress on intertidal habitats, particularly those sensitive to changes in submergence duration and salinity. In some cases, different components may be out of phase, leading to destructive interference where one contribution partially or completely cancels out another. This can moderate the extremes, but also adds complexity to predicting actual water levels. Understanding these interactions is essential for assessing the exposure and resilience of coastal habitats under current and future climate conditions.

To assess the influence of water level variability on intertidal habitats, we first identified a representative ENSO-neutral year using the Southern Oscillation Index (SOI) as a guiding metric. By analyzing the duration each year remained within the SOI range of -1 to +1, the year 2018 was selected as the most stable ENSO-neutral year during the study period. To simulate the impact of ENSO phases on water levels, we introduced two synthetic scenarios: an ENSO-positive case (+10 cm water level) and an ENSO-negative case (-10 cm water level).

Figure 32 presents the resulting differences in submergence percentage, comparing both ENSO-positive and ENSO-negative scenarios against the neutral baseline. As expected, the variations in submergence ranged from 10% to 20% across the domain. When mapped against existing mangrove areas, these differences were relatively minimal, suggesting mangroves are less sensitive to such water level shifts. In contrast, regions dominated by cyanobacterial mats exhibited more pronounced sensitivity, with noticeable differences in submergence.

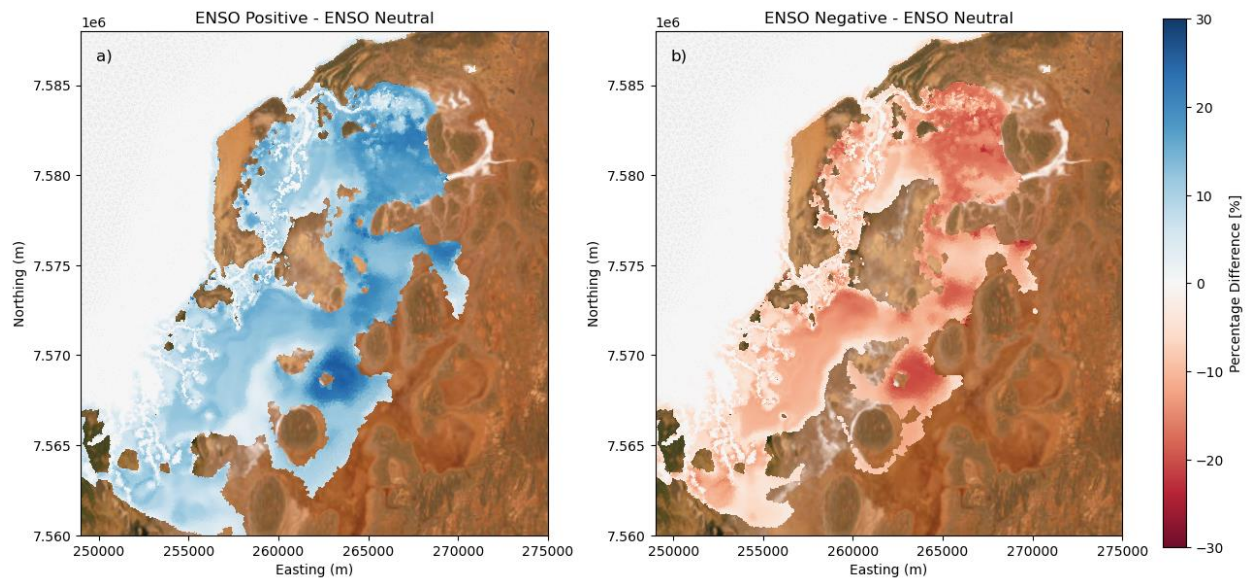


Figure 32 Difference in the percentage of submergence based on ENSO positive, negative, compared to ENSO neutral conditions.

To further investigate the spatial effects, we compared the inundation extents for both ENSO-positive and ENSO-negative scenarios separately (Figure 33). Overlaying these extents with existing habitat distributions revealed that high intertidal flats and salt flats lie within zones of greatest inundation variability. Cyanobacterial mat and mangrove habitats, while spanning wet and dry zones, were observed to be the most affected in terms of shifting exposure, particularly the cyanobacterial mats, which appear to respond more strongly to subtle hydrodynamic changes.

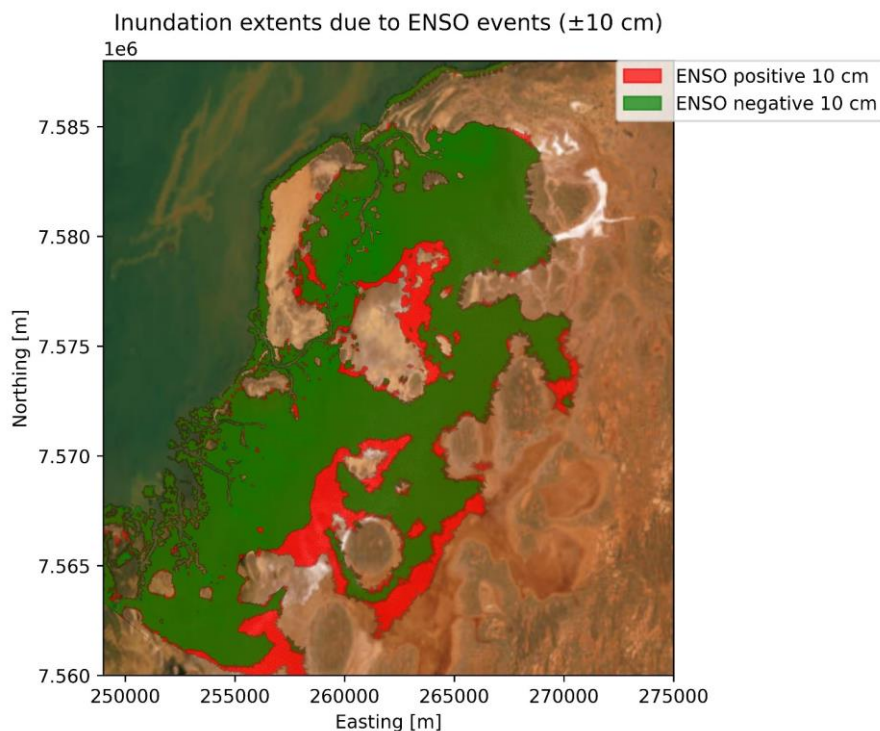


Figure 33 Inundation extent for ENSO positive and negative scenarios.

7. Limitations

While this study presents insight into the future distribution of intertidal habitats under various sea-level rise (SLR) scenarios, several limitations must be acknowledged, stemming from modelling assumptions, data availability, and natural system complexities.

One of the most significant limitations is the assumption of static creek geometries. In reality, tidal creeks are dynamic systems whose width, depth, and extent can evolve over time due to morphodynamic processes (i.e., resulting from the cumulative effect of changing sediment transport processes). These morphological changes can alter tidal flow pathways and, consequently, influence inundation extents, which may, in turn, impact intertidal habitat distributions. New mangrove colonization and survival are governed by complex interactions of physical and ecological processes across different stages of their life cycle, including seedling dispersal and establishment, anchoring and root development, and seedling growth and survival. The inland extension of creeks can facilitate the colonization of previously dry upland areas by mangroves, as it improves hydrological connectivity and expands the intertidal area available for propagule settlement. However, our current modelling approach assumes static bathymetry, which precludes the simulation of such creek evolution. As a result, the model may provide a conservative estimate of the inland extent of inundation, particularly under higher sea level scenarios projected for 2075 and 2100. This limitation restricts the predicted mangrove expansion into areas where, in reality, evolving creek networks could enable suitable conditions for colonization. That said, due to the flat nature of the elevation in the Urula region, it is also possible that the model overestimates mangrove suitability in some areas. These factors highlight the uncertainty associated with the prediction of future mangrove distribution. By not accounting for potential creek evolution, the model may underestimate or overestimate future inundation, leading to inaccuracies in the habitat predictions.

In addition to tidal creek evolution, the assumption of static bathymetry limits the model's ability to capture broader coastal morphological changes in response to sea-level rise (SLR). For instance, with higher SLR scenarios, the perimeter of the domain could expand as coastal dunes erode and salt flats migrate further inland. These processes are not currently represented in the model, potentially introducing further uncertainty, particularly under high SLR conditions. As a result, some habitats may not become permanently "drowned" as predicted, but instead could shift landward along elevated margins. This dynamic response is especially relevant in arid coastal systems where such topographic transitions can influence long-term habitat resilience and adaptation.

Moreover, the model does not account for potential feedback between habitat distribution and morphodynamics (eco-morphodynamic processes). For example, mangrove migration into new areas can alter sediment deposition, roughness, and local hydraulics, potentially reshaping channels or promoting sediment accretion. These eco-geomorphic feedbacks are complex and remain poorly understood, especially in arid coastal wetland systems like those of the Pilbara. While our approach focuses primarily on hydrodynamic drivers, future efforts could benefit from an integrated eco-morphodynamic framework to better represent these interactions and enhance the robustness of long-term habitat predictions.

Habitat predictions were made solely based on hydrodynamic conditions — namely, water depth, inundation duration, and exposure periods derived from the hydrodynamic model. While these parameters are critical for determining a habitat's "hydrological niche", they do not encompass other drivers that may influence habitat establishment, such as soil properties, ocean/atmospheric warming and ocean acidity or ecosystem feedbacks, which will also play roles in habitat establishment and resilience. However, they have been effective at explaining intertidal habitat distributions in other studies (e.g., Henderson & Glamore, 2024). This modelling approach thus simplifies the complex interplay of factors shaping intertidal ecosystems. In particular, root-zone salinity can be an important control on the landward extent of mangrove habitats (Wimmler et al., 2024), and it has not been explicitly included in this assessment. There is increasing evidence that both salinity and temperature gradients influence the physiological tolerance, growth, and distribution of many intertidal organisms, including both cyanobacterial mats and mangroves. However, over the time frames in question, the new inundation regimes that emerge will generate new equilibrium conditions in the salinity gradient that we assume will occur in tandem with the hydrological changes, allowing previous habitats that were too saline to become colonisable. Variations in freshwater input (rainfall in this case which is ignored for the future predictions), evaporation, and tidal mixing may cause other changes to the spatiotemporal variability in salinity and temperature, which are not captured in this model and we note the prediction framework does not fully reflect the multi-stressor environment experienced by intertidal habitats. Thresholds used to classify habitat suitability were defined using the 25th to 75th percentiles of habitat-specific hydrodynamic parameters. While this range captures central tendencies, it excludes potential outliers or extremes, possibly missing important habitat variability. Furthermore, the reference habitat maps used to extract these thresholds were only ~70% accurate, meaning up to 30% of the classification could be incorrect. Errors in the mapped data could propagate through the analysis and compromise the accuracy of future predictions.

Due to the lack of long-term and high-resolution field observations, the model predictions could only be validated against the existing mapped habitat data, which itself may contain errors. This creates a validation loop where predictions are tested against potentially flawed references, limiting confidence in model accuracy and transferability.

There is a notable discrepancy in the spatial resolution between the habitat maps (30 m × 30 m) and the hydrodynamic model grid (ranging from 12 m to 250 m). This resolution mismatch can introduce interpolation or generalization errors, particularly in areas where steep gradients in hydrodynamic conditions or habitat boundaries exist. For example, small creeks or narrow intertidal zones might be resolved in the model but generalized in the habitat data, leading to spatial misalignments in predictions.

The SLR scenarios used in the study are derived from multiple projected pathways (e.g., SSP scenarios) and therefore inherently carry uncertainty. The exact trajectory of SLR will depend on future global emissions, ice sheet dynamics, and regional oceanographic processes. If the underlying water level projections are inaccurate, then the downstream habitat predictions will also be unreliable.

The elevation and seabed data used in the model are key inputs for simulating water flow and inundation patterns. However, these data often come from various sources that are interpolated or remotely-sensed sources, which can contain errors e.g., due to vegetation cover, water

turbidity, or limited survey coverage. Inaccuracies in topography and bathymetry directly affect water routing and inundation timing, introducing further uncertainty into habitat predictions.

In this study, static roughness values were assigned based on the initial mapped habitat types and were not dynamically updated during the simulation. However, as habitats shift due to inundation or exposure, they can significantly alter flow resistance (e.g., dense mangrove roots increasing drag). The lack of dynamic feedback between habitat change and hydrodynamic conditions means the model may only be accurate in the short term and might deviate substantially over longer predictions.

Although storm events (e.g., TC Herman, TL 16U) are included in the model via observed water levels, their effects were not analysed separately. Storm surges can cause short-term but extreme inundation, which might temporarily affect habitat conditions (e.g., salinity shock, erosion). Including them in the mean conditions may skew the prediction thresholds, especially if the focus is on long-term habitat stability.

8. Summary and conclusions

This study integrates hydrodynamic modelling and data-driven habitat prediction to assess potential changes in intertidal habitat distributions under various projected future sea level scenarios. The Delft3D-FM hydrodynamic model, developed as part of the WAMSI Mardie Offset Research Project B, was applied to simulate water levels and currents across the study domain. These simulations provided the foundational input for a data-driven Random Forest model, which predicts habitat migration (using a habitat suitability index approach) based on historical inundation regimes and hydrodynamic conditions, such as submergence/exposure duration, water depth, and currents. A matrix of scenarios was developed to explore the sensitivity of habitat change and predict habitat suitability extents and variations under SLR scenarios with and without the salt development bunds, and highlight the influence of sea level anomalies, e.g., ENSO effects. These findings provide preliminary evidence for the likely extent of changes to the theoretical niches of intertidal habitats under the combined pressures of sea level rise and development.

The results indicate a tendency for intertidal habitats to shift inland in response to rising sea levels. As inundation extents move progressively inland, habitat types adapt to maintain alignment with their preferred inundation regime. Among these, cyanobacterial mats show the most notable changes, migrating inland as water levels rise and expanding their spatial footprint up to a threshold, beyond which deeper water is most likely to become limiting. Mangroves, by contrast, exhibit relatively stable distributions under moderate SLR but begin to experience loss and inland retreat under more extreme scenarios.

The inclusion of a development boundary analysis revealed that, while mangrove encroachment into this zone remains limited, a significant portion of cyanobacterial mat is predicted to migrate into the development area. This suggests that increasing habitat suitability within the zone for cyanobacterial mats is likely under future hydrodynamic conditions. Moreover, habitat expansion into previously unclassified areas—such as highland fringes and regions that were dry under baseline conditions—indicates that SLR may enable new zones of ecological colonisation,

particularly for salt flats and high intertidal flats (noting that high-intertidal flats also encompass areas with active cyanobacterial mat features).

When assessing the total area change across habitat types, cyanobacterial mats show the greatest variability, both in terms of footprint expansion and sensitivity to submergence duration. Mangroves demonstrate more conservative shifts, maintaining resilience to moderate water level changes but showing vulnerability under extreme conditions. Salt flats and high intertidal flats respond dynamically to shifts in the inundation regime, often emerging in transitional zones.

ENSO-related sea level anomalies further modulate these patterns. Periodic sea level changes of ± 10 cm, associated with ENSO phases, significantly alter submergence conditions across the domain. While mangrove habitats remain largely stable under such variability, cyanobacterial mats respond strongly to even minor changes in water depth and exposure duration. These preliminary findings highlight the importance of considering both long-term SLR and shorter-term sea level anomalies when evaluating habitat resilience and future migration potential.

In summary, the integrated modelling framework provides valuable insights into habitat dynamics under multiple stressors, supporting informed decision-making for coastal development, conservation planning, and future ecological risk assessments.

9. Appendix A

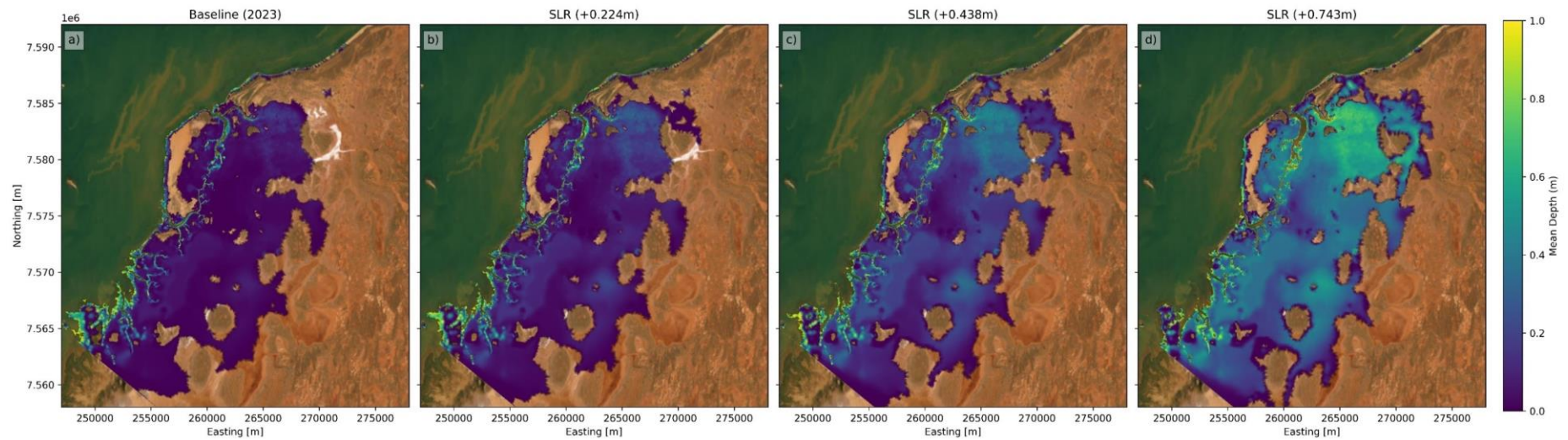


Figure A 1 Mean water depth across different SLR scenarios without the development boundary.

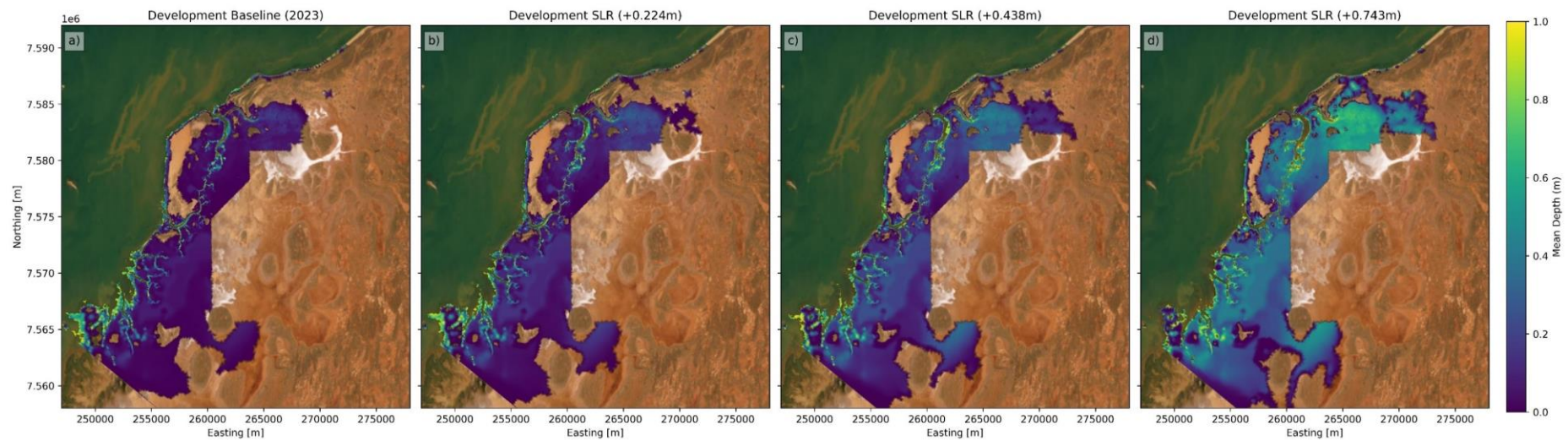


Figure A 2 Mean water depth across different SLR scenarios with the development boundary.

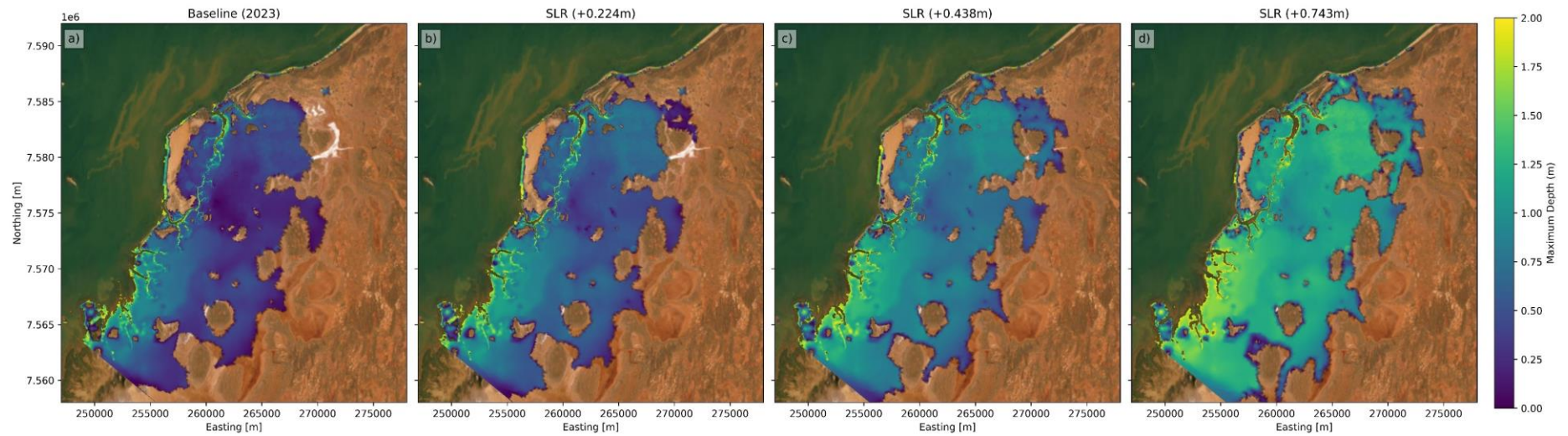


Figure A 3 Maximum depth across different SLR scenarios without the development boundary.

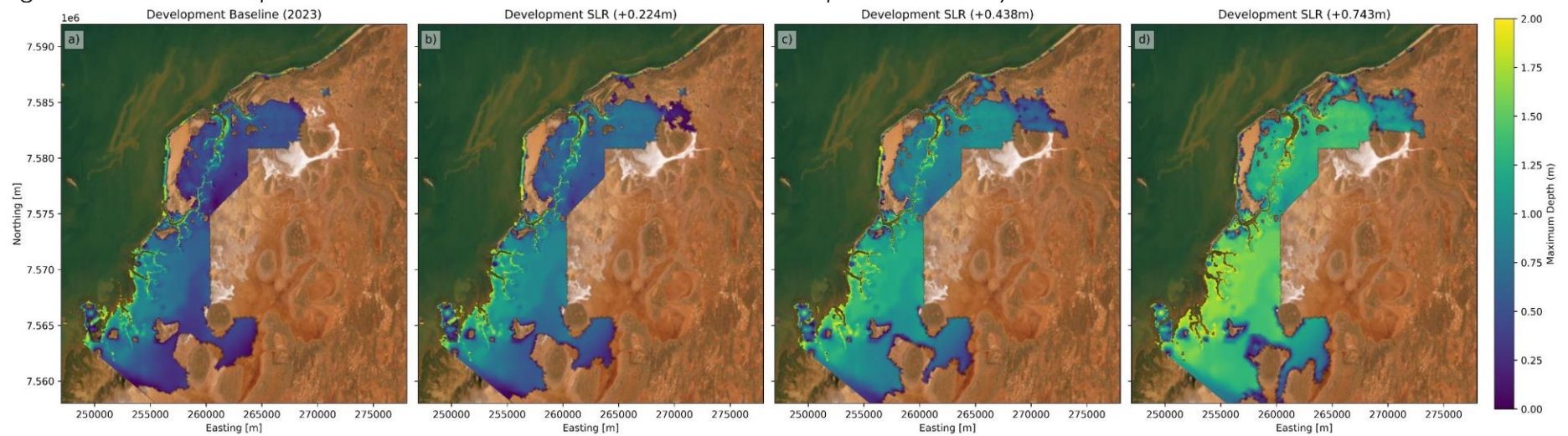


Figure A 4 Maximum depth across different SLR scenarios with the development boundary.

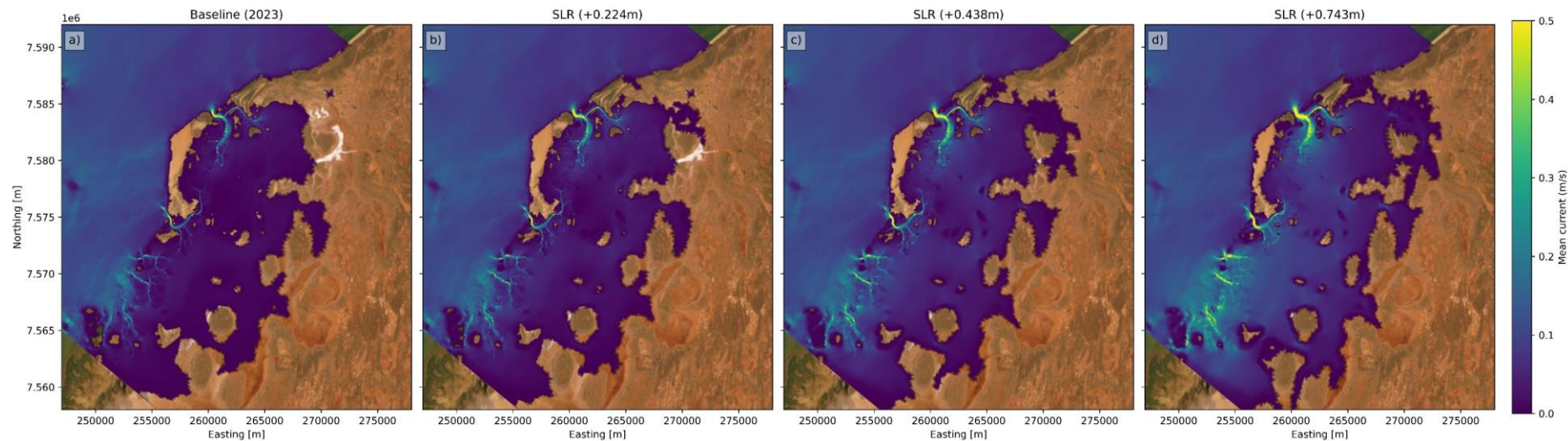


Figure A 5 Mean current magnitude across different SLR scenarios without the development boundary

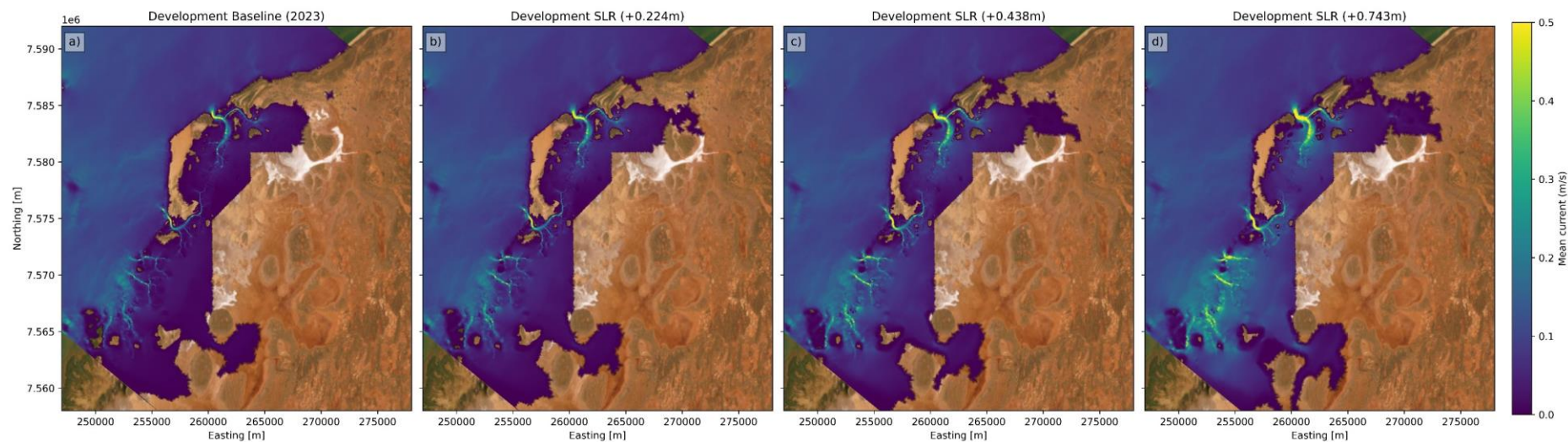


Figure A 6 Mean current magnitude across different SLR scenarios with the development boundary.

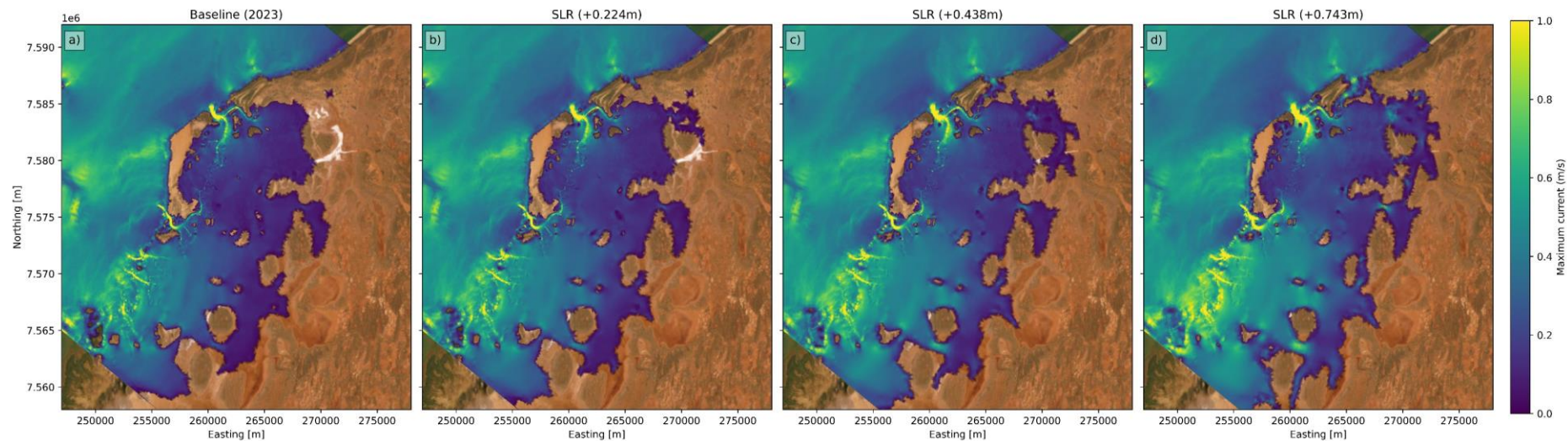


Figure A 7 Maximum current magnitude across different SLR scenarios without the development boundary.

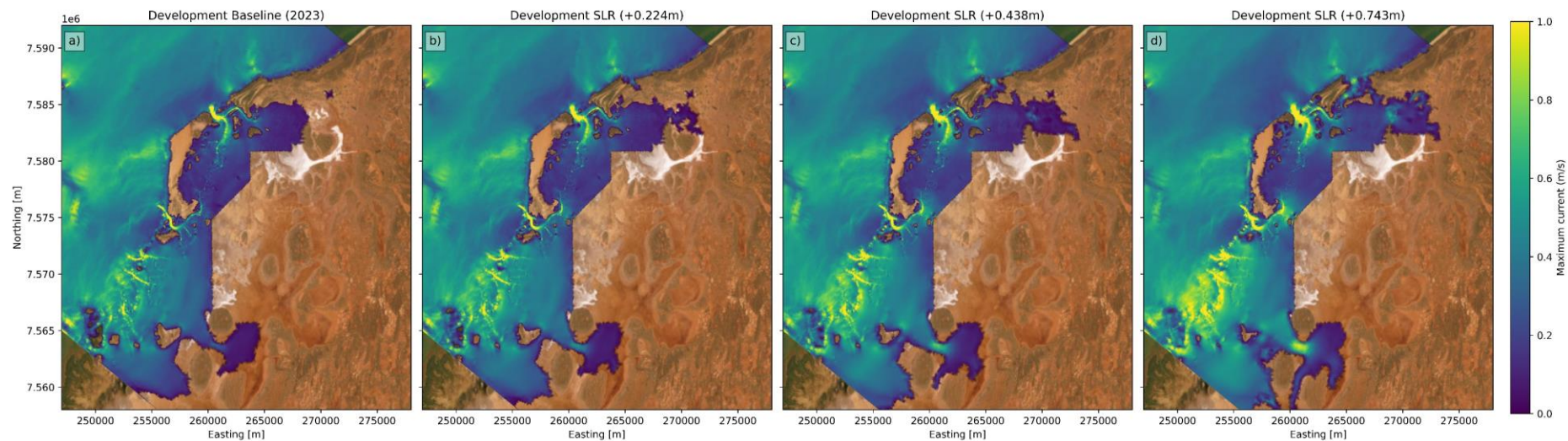


Figure A 8 Maximum current magnitude across different SLR scenarios with the development boundary.

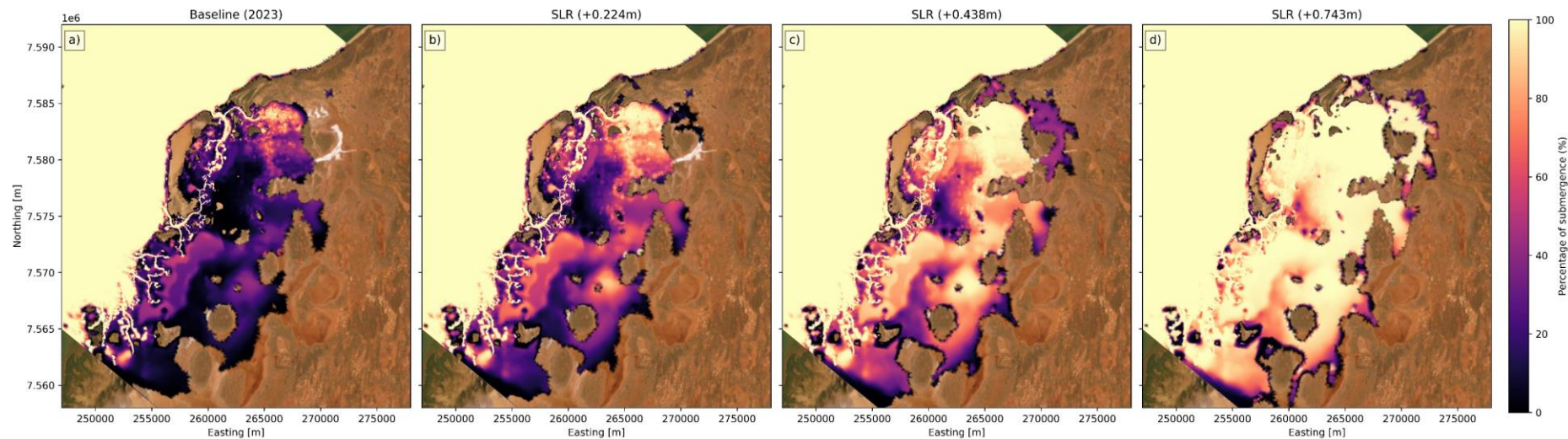


Figure A 9 Percentage of submergence based on 1-year simulation for different SLR scenarios without the development boundary.

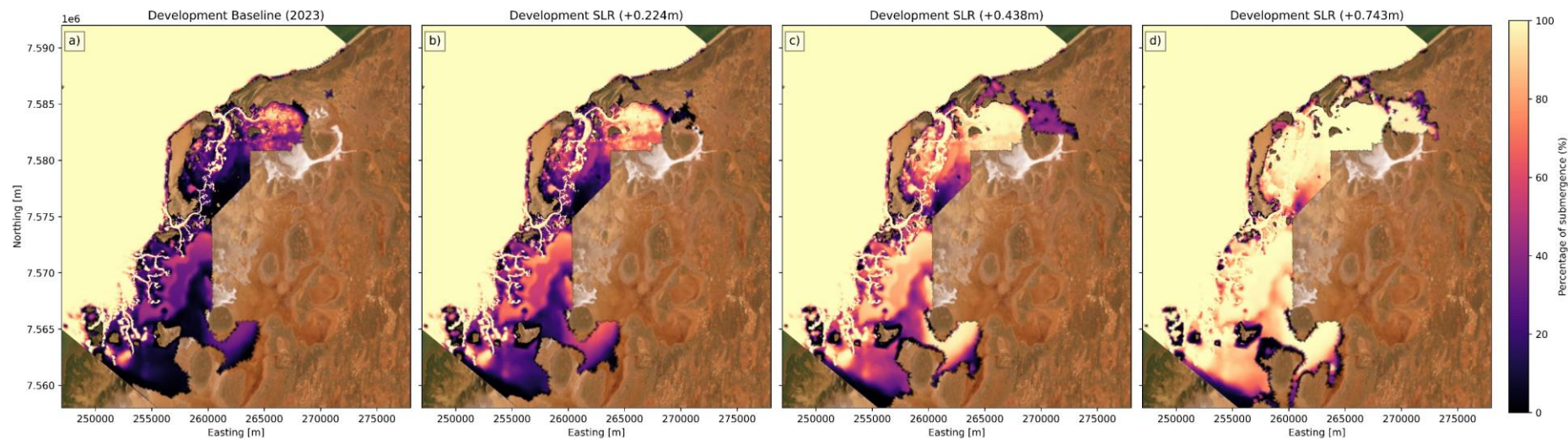


Figure A 10 Percentage of submergence based on 1-year simulation for different SLR scenarios with the development boundary.

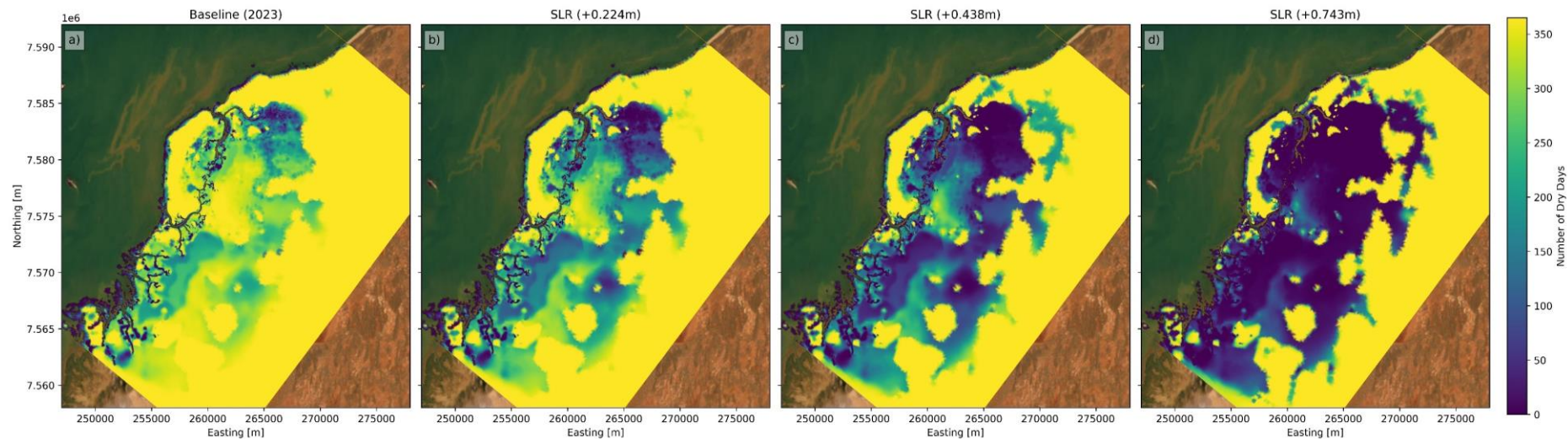


Figure A 11 Number of dry days for different SLR scenarios without the development boundary.

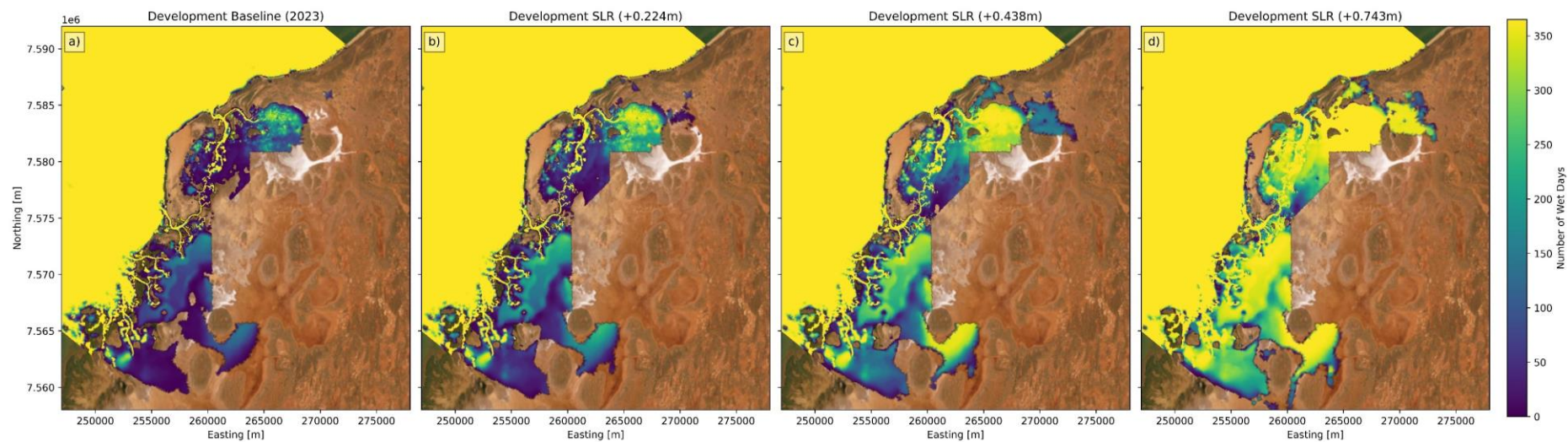


Figure A 12 Number of dry days for different SLR scenarios with the development boundary.

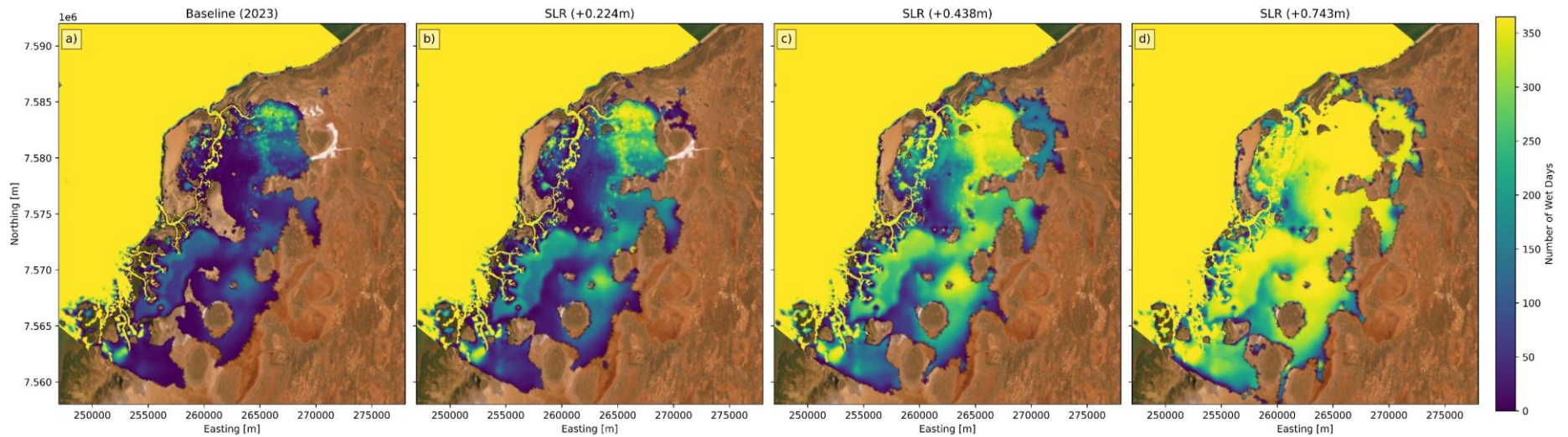


Figure A 13 Number of wet days across different SLR without the development boundary.

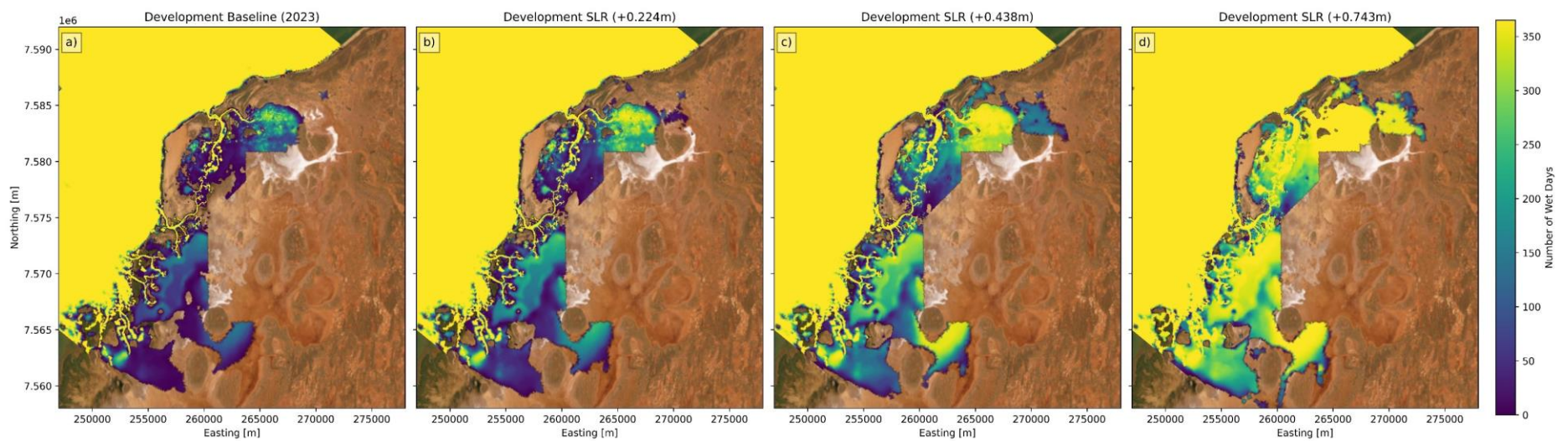


Figure A 14 Number of wet days across different SLR with the development boundary.

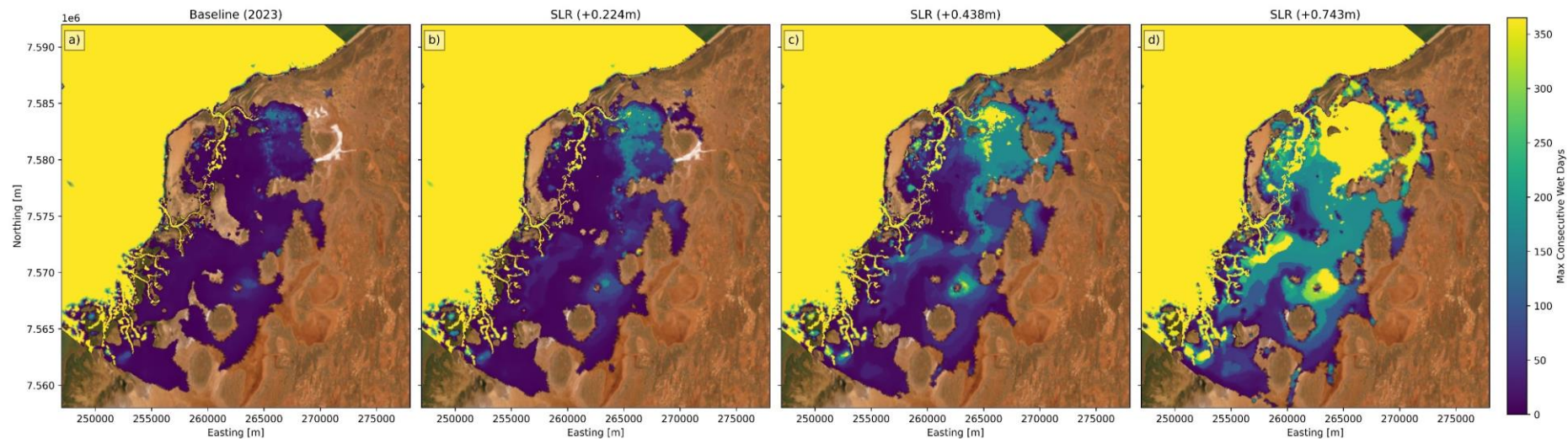


Figure A 15 Maximum consecutive wet days across different SLR scenarios without the development boundary.

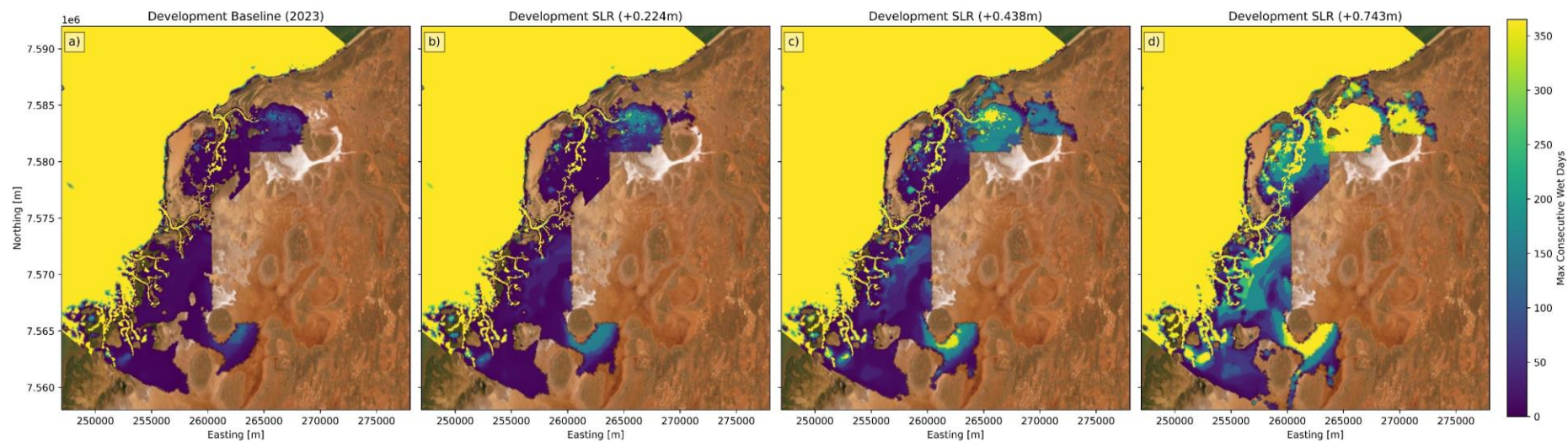


Figure A 16 Maximum consecutive wet days across different SLR scenarios with the development boundary.

10. References

- Adame, M. F., Reef, R., Santini, N. S., Najera, E., Turschwell, M. P., Hayes, M. A., Masque, P., & Lovelock, C. E. (2021). Mangroves in arid regions: Ecology, threats, and opportunities. *Estuarine, Coastal and Shelf Science*, 248. <https://doi.org/10.1016/j.ecss.2020.106796>
- Allen, M. R., O.P. Dube, Solecki, W., Aragón-Durand, F., Cramer, W., Humphreys, S., Kainuma, M., Kala, J., Mahowald, N., Mulugetta, Y., Perez, R., Wairiu, M., & Zickfeld, K. (2018). IPCC: Framing and Context. In *Global warming of 1.5°C. An IPCC Special Report*.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324/METRICS>
- Calvin, K., Dasgupta, D., Krinner, G., Mukherji, A., Thorne, P. W., Trisos, C., Romero, J., Aldunce, P., Barrett, K., Blanco, G., Cheung, W. W. L., Connors, S., Denton, F., Diongue-Niang, A., Dodman, D., Garschagen, M., Geden, O., Hayward, B., Jones, C., ... Ha, M. (2023). *IPCC, 2023: Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team, H. Lee and J. Romero (eds.)]. IPCC, Geneva, Switzerland.* (P. Arias, M. Bustamante, I. Elgizouli, G. Flato, M. Howden, C. Méndez-Vallejo, J. J. Pereira, R. Pichs-Madruga, S. K. Rose, Y. Saheb, R. Sánchez Rodríguez, D. Ürge-Vorsatz, C. Xiao, N. Yassaa, J. Romero, J. Kim, E. F. Haites, Y. Jung, R. Stavins, ... C. Péan, Eds.). <https://doi.org/10.59327/IPCC/AR6-9789291691647>
- Cannon, D., Kibler, K., Donnelly, M., McClenachan, G., Walters, L., Roddenberry, A., & Phagan, J. (2020). Hydrodynamic habitat thresholds for mangrove vegetation on the shorelines of a microtidal estuarine lagoon. *Ecological Engineering*, 158, 106070. <https://doi.org/10.1016/J.ECOLENG.2020.106070>
- Demoulin, C. F., Lara, Y. J., Cornet, L., François, C., Baurain, D., Wilmotte, A., & Javaux, E. J. (2019). Cyanobacteria evolution: Insight from the fossil record. *Free Radical Biology and Medicine*, 140, 206–223. <https://doi.org/10.1016/J.FREERADBIOMED.2019.05.007>
- Hague, B. S., McGregor, S., Murphy, B. F., Reef, R., & Jones, D. A. (2020). Sea Level Rise Driving Increasingly Predictable Coastal Inundation in Sydney, Australia. *Earth's Future*, 8(9), e2020EF001607. <https://doi.org/10.1029/2020EF001607>; WEBSITE: WEBSITE: AGUPUBS; WGROU: STRING: PUBLICATION
- Henderson, B., & Glamore, W. (2024). A lifecycle model approach for predicting mangrove extent. *Science of The Total Environment*, 952, 175962. <https://doi.org/10.1016/J.SCITOTENV.2024.175962>
- Hickey, S., Stone, A., & Lovelock, C. (2023). *The Cyanobacterial Mats of the Exmouth Gulf, Western Australia: Mapping Report*. <https://doi.org/10.26182/JJIV-4R98>
- Kernkamp, H. W. J., Van Dam, A., Stelling, G. S., & De Goede, E. D. (2011). Efficient scheme for the shallow water equations on unstructured grids with application to the Continental Shelf. *Ocean Dynamics*, 61(8), 1175–1188. <https://doi.org/10.1007/S10236-011-0423-6/TABLES/2>

- Kumbier, K., Hughes, M. G., Rogers, K., & Woodroffe, C. D. (2021). Inundation characteristics of mangrove and saltmarsh in micro-tidal estuaries. *Estuarine, Coastal and Shelf Science*, 261, 107553. <https://doi.org/10.1016/J.ECSS.2021.107553>
- Lovelock, C. E., Feller, I. C., Reef, R., Hickey, S., & Ball, M. C. (2017). Mangrove dieback during fluctuating sea levels. *Scientific Reports*, 7(1). <https://doi.org/10.1038/s41598-017-01927-6>
- Lowe, R. J., Cuttler, M. V. W., & Hansen, J. E. (2021). Climatic Drivers of Extreme Sea Level Events Along the Coastline of Western Australia. *Earth's Future*, 9(4). <https://doi.org/10.1029/2020EF001620>
- Mazda, Y., Magi, M., Nanao, H., Kogo, M., Miyagi, T., Kanazawa, N., & Kobashi, D. (2002). Coastal erosion due to long-term human impact on mangrove forests. *Wetlands Ecol Manage*, 10(10), 1–9. <https://doi.org/10.1023/a:1014343017416>
- Merrifield, M. A., Johnson, M., Guza, R. T., Fiedler, J. W., Young, A. P., Henderson, C. S., Lange, A. M. Z., O'Reilly, W. C., Ludka, B. C., Okihira, M., Gallien, T., Pappas, K., Engeman, L., Behrens, J., & Terrill, E. (2021). An early warning system for wave-driven coastal flooding at Imperial Beach, CA. *Natural Hazards*, 108(3), 2591–2612. <https://doi.org/10.1007/s11069-021-04790-x>
- Oliveira, S., Oehler, F., San-Miguel-Ayanz, J., Camia, A., & Pereira, J. M. C. (2012). Modeling spatial patterns of fire occurrence in Mediterranean Europe using Multiple Regression and Random Forest. *Forest Ecology and Management*, 275, 117–129. <https://doi.org/10.1016/J.FORECO.2012.03.003>
- Rybak, A. S., Dziuba, M., Pełechata, A., Rybak, M., Akter, S., Czerepska, A., Dulić, T., Gąbka, M., Hindáková, A., Jurczak, T., Kendir, A., Mankiewicz-Boczek, J., Meriluoto, J., & Wejnerowski, Ł. (2024). Characterization of cyanobacterial mats from an artificial hot spring in Uniejów (Poland) and the potential use of their biomass. *Algal Research*, 82, 103646. <https://doi.org/10.1016/J.ALGAL.2024.103646>
- Vrieling, A., Meroni, M., Darvishzadeh, R., Skidmore, A. K., Wang, T., Zurita-Milla, R., Oosterbeek, K., O'Connor, B., & Paganini, M. (2018). Vegetation phenology from Sentinel-2 and field cameras for a Dutch barrier island. *Remote Sensing of Environment*, 215, 517–529. <https://doi.org/10.1016/J.RSE.2018.03.014>
- Wimmler, M. C., Bathmann, J., Peters, R., Jiang, J., Walther, M., Lovelock, C. E., & Berger, U. (2021). Plant–soil feedbacks in mangrove ecosystems: establishing links between empirical and modelling studies. *Trees* 2021 35:5, 35(5), 1423–1438. <https://doi.org/10.1007/S00468-021-02182-Z>
- Wimmler, M. C., Bathmann, J., Vollhüter, J., & Berger, U. (2024). pyMANGA: A modular, open and extendable software platform for modeling of forest and vegetation dynamics. *Environmental Modelling & Software*, 174, 105973. <https://doi.org/10.1016/J.ENVSOFT.2024.105973>
- Xie, D., Schwarz, C., Kleinhans, M. G., Zhou, Z., & van Maanen, B. (2022). Implications of Coastal Conditions and Sea-Level Rise on Mangrove Vulnerability: A Bio-Morphodynamic Modeling

Study. *Journal of Geophysical Research. Earth Surface*, 127(3).
<https://doi.org/10.1029/2021JF006301>

